

HJ-Gauss

Monte-Carlo Hamilton-Jacobi Reachability

Full Exposition: From TVD-RK Levelsets to Monte Carlo Sampling for BRTs.

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BLUF (Bottom Line Up Front) — 1/2

Grid-resolved Hamilton-Jacobi (HJ) BRTs yield certified safety sets at a $O(M^n)$ memory cost.

-  **Background:** LevelSetPy resolves backward reachable sets/tubes by discretizing the HJ PDE (upwinding + Lax-Friedrichs + TVD-RK) on a grid — certified, but exponential in state dimension n .
-  **Idea:** a Cole-Hopf transformation turns the *viscous* HJ PDE into a **linear heat equation**; Feynman-Kac evaluates it as a **Gaussian expectation**. No grid.

HJ-Gauss replaces the grid with Gaussian Monte-Carlo, dropping the footprint to $O(N \cdot n)$.

BLUF (Bottom Line Up Front) — 2/2

HJ-Gauss replaces the grid with Gaussian Monte-Carlo, dropping the footprint to $O(N \cdot n)$.

-  **Generality:** a frozen-coefficient **Picard iteration** handles the nonconvex, state-dependent Hamiltonians of reachability.
-  **Rigor:** $O(N^{-1/2})$ concentration, contraction convergence, a linearization residual bound, and a **conservative safety certificate**.
-  **For this room:** the windowed reachable tube is a dynamics-aware **conflict predicate** for MAPF; the value gradient is a **score** for diffusion planners.

This Gaussian kernel is the diffusion kernel, the bridge to learning-based planners.

Roadmap — Main Talk (Parts 1-5)

Part	Theme
1	The wall: curse of dimensionality & prior grid-free work
2	HJ-Gauss core theory: Cole-Hopf → heat → Feynman-Kac → Picard
3	Importance sampling & variance control
4	Guarantees: concentration, contraction, residual, certificate
5	Experiments (rockets, Dubins, 45D, 10^5 birds)

 **Continue:** Parts 6-9 on the next slide.

Roadmap — Main Talk (Parts 6-9) & Appendix

Part	Theme
6	Application: multi-agent path finding (AMFS)
7	The diffusion connection
8	Dirty Laundry: limits & boundaries
9	Conclusions, future work, references
 A–C	Appendix: HJ/viscosity theory (A) · reachability foundations (B) · LevelSetPy grid pipeline (C)

The background — **HJ PDE and viscosity theory (A)**, **Reachability Foundations (B)**, and the **LevelSetPy grid pipeline (C)** — is collected in the **Appendix**.

Notations

Symbol	Meaning
$x \in \Omega \subseteq \mathbb{R}^n$	x : State; Ω : Open set; n = State dimension.
$v(t, x)$	Value function; v_t time derivative; $Dv = \nabla_x v$ spatial gradient (co-state).
$H(t; x, p)$	Hamiltonian; p = co-state.
$g(x)$	Terminal/target datum (signed distance $\ell(x)$); BUC.
$\delta > 0$	Viscosity parameter.

Notations

Symbol	Meaning
$\omega^\delta = e^{-cv^\delta}$	Cole-Hopf transformed variable.
$c(t; x)$	Frozen coefficient $= \frac{2}{\delta} H^\delta / Dv^\delta ^2$.
M, N	Grid points per dimension (M); Monte-Carlo samples per query state (N).
$\mathcal{L}_0, \mathcal{L}$	Target set, backward reachable tube (BRT).

Two independent counts: M evaluation states (arbitrary, grid-free) vs N Gaussian samples drawn *per state*. Total randomness per iteration $= M \times N$.

Part 1

The Wall

Curse of dimensionality, and who has tried to climb it.

The Curse of Dimensionality, Quantified

Grid memory is $O(M^n)$ with M points per dimension:

State dim n	$M = 100$ cells	Feasibility (double array)
2	10^4	✓ trivial.
3	10^6	✓ easy.
4	10^8	⚠ ~0.8 GB, heavy.
5	10^{10}	✗ ~80 GB.
6	10^{12}	✗ ~8 TB.
45 (our multi-agent game)	10^{90}	✗ exceeds atoms in the observable universe.

- GPU acceleration lowers wall-clock, **not** the exponent.

Reachability's most valuable regime — many interacting agents, rich dynamics — is where grids are impossible.



Attempt 1: PINN Reachability (DeepReach)

- **DeepReach** trains a neural network to minimize the HJ PDE residual directly (physics-informed loss), no reference grid.
- **Pros:** Trades grid storage for network weights; sidesteps $O(M^n)$ memory.
- **Con:** Scales only to moderate dimensiond (reported ~9-10D);
 - And **accuracy degrades** as n grows; training is a nonconvex optimization with no viscosity-solution guarantee.

Catch-22: Learning-based solvers help, but do not deliver a certified, dimension-robust representation.
HJ-Gauss keeps the viscosity-solution semantics.



Attempt 2: Convex-Duality Formulas (Hopf / Lax-Oleinik)

HJ-Gauss trades their exactness for generality, at the quantified price of a linearization residual bound (Part 4).

- **Hopf** and **Lax-Oleinik** formulas¹ evaluate the value **pointwise** via convex optimization — grid-free at a point.
- **Limitation:** they require a **convex** (or state-independent) Hamiltonian.
- Reachability's $H(x, p) = \max_u \min_w \langle p, f \rangle$ is **state-dependent and nonconvex** → outside this class.

It handles the nonconvex, state-dependent Hamiltonian head-on.

¹ Darbon-Osher 2016; Chow-Darbon-Osher-Yin 2017; Kirchner 2018

Attempts 3-4: Decomposition, Path Integral, Sampling

Family	Idea	Boundary
System decomposition ¹	Remove dimension by splitting self-contained subsystems	Only when coupling structure permits; HJ-Gauss is indifferent to coupling and <i>complements</i> it.
Stochastic PDE / path integral ²	Log-substitution linearizes the stochastic HJB under noise-control duality	Diffusion comes from <i>process noise</i> ; ours is a deterministic worst-case game , δ an analysis parameter.
Sample-based stochastic reachability ³	Certify probabilistic reach-avoid	We certify adversarial worst-case reachability — a different, stronger guarantee.

The open gap: a grid-free, storage-light scheme for **nonconvex, state-dependent, adversarial** Hamiltonians. That is HJ-Gauss.

¹ Chen-Herbert · ² Kappen; Theodorou · ³ Summers-Lygeros; Lesser-Oishi



Where HJ-Gauss Sits

- Keeps the **viscosity-solution** semantics (unlike PINN heuristics).
- Handles **nonconvex, state-dependent** H (unlike Hopf/Lax-Oleinik).
- **Indifferent to coupling structure** (complements decomposition).
- Certifies **adversarial worst-case** safety (unlike probabilistic sampling).
- Memory $O(N \cdot n)$, grid- and discretization-free.

The rest is *how*: a Cole-Hopf transformation, a Feynman-Kac expectation, and a frozen-coefficient Picard iteration.

Part 2

HJ-Gauss Core Theory

Cole-Hopf Transformation $\rightarrow$ Linear Heat Equation



$\rightarrow$ Feynman-Kac Formula $\rightarrow$ Picard Iteration.

The Theoretical Roadmap of the Method

1. Start from the **viscous** HJ PDE (Part A).
2. Apply a generalized **Cole-Hopf** transform $\omega = e^{-cv} \rightarrow$ a **Linear heat equation**.
3. Solve the heat equation by its **Gaussian heat kernel** $\rightarrow$ a **Feynman-Kac expectation**.
4. Recover value (log-sum-exp) and gradient from that expectation.
5. Because c depends on the unknown Dv , **iterate**: freeze, solve, update (Picard) = **Algorithm 1**.
6. Analyze exactness (quadratic case) vs the **quasi-linearization residual** (general case).

Each arrow is a theorem or proposition in the paper; we walk them one by one.

The Transformation, and the Closed-Form/Approximate Split

For a general H , define the spatially-varying coefficient

$$c(t; x) = \frac{2}{\delta} \cdot \frac{H^\delta}{|Dv^\delta|^2}, \quad H^\delta := H(t; x, Dv^\delta),$$

and set $\omega^\delta := \exp(-c v^\delta)$.

- **Quadratic case** $H = \frac{1}{2}|p|^2$: $c = 1/\delta$ constant, ω^δ solves the **homogeneous heat equation** with **zero residual** — a genuine Cole-Hopf identity.
- **General H** : the transform induces a residual $R = R_{\text{alg}} + R_{\text{der}}$;
 - c 's choice eliminates the **algebraic** part R_{alg} , leaving the **derivative** part R_{der} .

No freezing of c 's *values* removes its *derivatives*. Hence for nonlinear H this is a **quasi-linearization**, not an identity — and the leftover R_{der} is the residual we will bound.

Picard Quasi-Linearization: The Surrogate View

- At iteration k , **freeze** $c^{(k)}$ at the current iterate and solve the linear heat equation **in closed form**.
- Equivalently: replace the true Hamiltonian by the **locally-matched quadratic surrogate**

$$\tilde{H}^{(k)}(p) = \frac{\delta}{2} c^{(k)} |p|^2,$$

solve the resulting viscous HJ equation in closed form, obtain $v^{(k+1)}$, update $Dv^{(k+1)}$ and $c^{(k+1)}$.

- **The iteration's limit is the fixed point of the surrogate solve map** — equal to the viscous solution *only* when $R_{\text{der}} \equiv 0$ (the quadratic case).

"Successive quadratic matching": each step is a locally closed-form quadratic-Hamiltonian solve, refined by re-estimating the co-state.

Reduction to the Heat Equation (Proposition)

With $c^{(k)}$ frozen, ω^δ solves the heat initial-value problem

$$\omega_t^\delta - \frac{\delta}{2} \Delta \omega^\delta = 0 \text{ in } \mathbb{R}^n \times (0, T], \quad \omega^\delta(0, x) = \exp(-c g(x)).$$

- When $H = \frac{1}{2}|p|^2$, $c = 1/\delta$: **closed-form, no residual**.
- For general H : discards the derivative residual (Lemma, appendix) — the quasi-linearization.

A nonlinear, nonconvex, state-dependent HJ PDE has become **the heat equation** — the most classical linear parabolic PDE, with an explicit Gaussian solution.

The Gaussian Heat Kernel & Feynman-Kac

The heat IVP has the unique bounded Green's-convolution solution

$$\omega^\delta(t, x) = \frac{1}{(\sqrt{2\pi\delta t})^n} \int_{\mathbb{R}^n} e^{-\frac{|x-y|^2}{2\delta t}} e^{-c g(y)} dy,$$

which, by **Feynman-Kac**, is a **Gaussian expectation**:

$$\omega^\delta(t, x) = \mathbb{E}_{y \sim \mathcal{N}(x, \delta t I_n)} [e^{-c g(y)}].$$

- The free-space kernel integrates to one over $\mathbb{R}^n$ (and no proper subset) — every estimator is posed on all of $\mathbb{R}^n$.
- Sampling $y \sim \mathcal{N}(x, \delta t I_n)$ and averaging $e^{-c g(y)}$ is an **unbiased** estimator of ω^δ .

The solution is a **Gaussian roll-out average of the terminal cost**. This is the entire computational payload — no grid, no marching.

Recovering the Value (Lemma)

Invert the transform:

$$v^\delta(t, x) = -\frac{1}{c^{(k)}} \log \mathbb{E}_{y \sim \mathcal{N}(x, \delta t I_n)} [e^{-c^{(k)} g(y)}].$$

- A **log-sum-exp** of Gaussian-sampled terminal costs.
- Numerically evaluated in a **log-domain-stable** form (log-mean-exp) to avoid overflow/underflow.
- M arbitrary evaluation states $x_1, \dots, x_M$ — laid on a grid or scattered — each with its own fresh N samples.

The value at a query point is a scalar reduction over N samples. Embarrassingly parallel across both M states and N samples.

Recovering the Gradient (Corollary)

The spatial gradient (co-state) has a closed Gaussian-expectation form:

$$Dv^\delta = \frac{1}{t \delta c^{(k)}} \left(x - \frac{\mathbb{E}_{y \sim \mathcal{N}(x, \delta t I_n)}[y e^{-c^{(k)} g(y)}]}{\mathbb{E}_{y \sim \mathcal{N}(x, \delta t I_n)}[e^{-c^{(k)} g(y)}]} \right).$$

- A **weighted-mean shift**: the gradient points from x toward the exponentially-weighted centroid of the samples.
- No finite differencing, no stored field — the co-state is a byproduct of the same samples.

This gradient is (up to constants) a **score function** $\nabla \log \omega$. Hold that: it is the entire bridge to diffusion in Part 7.

Admissible Data & Unbiasedness (Recap as a Guarantee)

- Datum $\omega^\delta(0, \cdot) = e^{-cg}$ is continuous, strictly positive, bounded above by $e^{-cg_{\min}}$, decaying at infinity.
- $\Rightarrow$ the heat solution is the **unique bounded** one; the integral converges absolutely; the Gaussian expectation is **genuine and unbiased**.
- Only $g \geq g_{\min}$ is required (bounded target); no upper bound on g .

The estimator drawing $y \sim \mathcal{N}(x, \delta t I_n)$ over all of $\mathbb{R}^n$ is **unbiased for the quantity inside the logarithm** — the foundation for the concentration bound of Part 4.

C Sign and the Removable Zero of the Coefficient

- $c \sim C_H / (\delta |Dv^\delta|)$: it **grows without bound** in flat regions and **vanishes and flips sign** where optimal dynamics run tangent to the level set (the barrier).
- The zero of c is a **removable** singularity of the estimator: as $c \rightarrow 0$,

$$-\frac{1}{c} \log \mathbb{E}[e^{-cg}] \rightarrow \mathbb{E}[g],$$

i.e. the log-sum-exp gracefully degenerates to the **Gaussian mean** of the terminal data — the correct pure-diffusion limit ($H \rightarrow 0$ reduces the PDE to the heat equation).

- In implementation: evaluate in a **log1p**-stable form and **clip** $|c|$ to $[c_{\min}, c_{\max}]$.

The coefficient's pathologies are analytically understood and numerically tamed — clipping is itself a bounded, priced perturbation (next slide).

Coefficient Regularization (Lemma)

Run Algorithm 1 with the **regularized coefficient**

$$c_\eta(t; x) = \frac{2}{\delta} \cdot \frac{H^\delta}{|Dv^\delta|^2 + \eta},$$

equivalent to running it unregularized for the **perturbed Hamiltonian** $H_\eta = H |Dv^\delta|^2 / (|Dv^\delta|^2 + \eta)$, which satisfies a **uniform** bound

$$\|H_\eta - H\|_\infty \leq \frac{C_H \sqrt{\eta}}{2},$$

so the induced value-function perturbation is at most $TC_H \sqrt{\eta}/2$.

Regularization cures the flat-interior blow-up at a **quantified, uniform** price — no hidden instability.

Algorithm 1: Quasi-Linearization (Cole-Hopf)

$$\text{Fix } \varepsilon > 0, \quad v^{(0)}(t, x) = g(x), \quad c^{(0)} = \frac{2 H(t, x, Dg)}{\delta |Dg|^2}.$$

for $k = 0, 1, 2, \dots$:

(1) freeze $c^{(k)}$ at the current iterate;

(2) solve $\omega_t = \frac{\delta}{2} \Delta \omega$, $\omega^{(k)}(0, x) = e^{-c^{(k)} g(x)}$;

$$\Rightarrow \omega^{(k+1)}(t, x) = \mathbb{E}_{y \sim \mathcal{N}(x, \delta t I)} [e^{-c^{(k)} g(y)}] \quad (\text{Monte-Carlo});$$

(3) recover $v^{(k+1)} = -\frac{1}{c^{(k)}} \log \omega^{(k+1)}$;

(4) update $Dv^{(k+1)}$, $c^{(k+1)} = \frac{2 H(t, x, Dv^{(k+1)})}{\delta |Dv^{(k+1)}|^2}$ clip to $[c_{\min}, c_{\max}]$;

(5) stop when $\|v^{(k+1)} - v^{(k)}\| / \|v^{(k)}\| < \varepsilon$.

- Steps (2)-(4) are all **Gaussian-expectation reductions** over N samples.
- Convergence typically in ≤ 20 iterations (empirically 12-15).

Every line is grid-free. The only state carried between iterations is the coefficient field $c^{(k)}$ at the M evaluation states.

Evaluation States vs Monte-Carlo Samples (Remark)

- N = i.i.d. Gaussian draws $y_1, \dots, y_N \sim \mathcal{N}(x, \delta t I_n)$ used **independently at each** evaluation state x to form the estimators.
- Resampled **fresh** at every evaluation state and every Picard iteration.
- M = the evaluation states $x_1, \dots, x_M$ where v is queried — carry **no randomness**, may be a uniform grid or scattered arbitrarily.
- Total randomness per iteration: $M \times N$ **draws**.
- The zero level set is recovered by evaluating v^δ at the M states and isocontouring at level zero (marching cubes) — a deterministic post-process consuming **no further samples**.

The **solve is grid-free**; the grids that appear in results are *evaluation windows* for visualization and error measurement, not sampling constraints.

Memory: $O(N \cdot n)$ vs $O(M^n)$

	Grid level sets	HJ-Gauss
Representation	Stored field on M^n grid	N Gaussian samples per query, n -vectors
Memory	$O(M^n)$	$O(N \cdot n)$
Discretization	Cartesian grid + marching cubes to solve	None (grid-free solve)
Query states	Tied to grid	Arbitrary / scattered
Parallelism	Grid array ops	Over M states and N samples

- Example: $N = 14,000$, $n = 3 \rightarrow \approx 0.6$ MB/iteration; $n = 45 \rightarrow \approx 7.2$ MB/iteration (vs 10^{90} grid cells).

Memory now scales with **sample budget and dimension**, not with a grid raised to the dimension. This is the whole contribution in one line.

Part 3

Importance Sampling & Variance Control

Making the Gaussian estimator work in high dimensions.

The Weight-Degeneracy Problem

- The estimators are a **ratio of exponential-weight expectations**, e.g. the gradient's numerator/denominator both carry e^{-cg} .
- In high dimensions (or small δ), $c g$ is large over the kernel support, so the denominator $\mathbb{E}[e^{-cg}]$ is carried by a **rare event**.
- The weights collapse onto a few samples $\rightarrow$ the ratio's **variance explodes**; the effective sample size crashes.

Naive Gaussian sampling is unbiased but can be catastrophically high-variance right where we need it (high n , small δ). We fix it with importance sampling.

Tilted Proposal & Density-Ratio Reweighting

- Replace $y \sim \mathcal{N}(x, \sigma^2 I_n)$ ($\sigma^2 = \delta t$) by a **shifted** proposal $q_\theta = \mathcal{N}(x + \theta, \sigma^2 I_n)$.
- Reweight by the **density ratio**

$$w_\theta(y) = \exp\left(\frac{|\theta|^2 - 2\langle y - x, \theta \rangle}{2\sigma^2}\right),$$

which leaves **every expectation unbiased**.

We move the samples to where the integrand has mass, then correct via the ratio. Bias is untouched; variance can collapse by orders of magnitude.

The Zero-Variance Shift Is One Gradient Step

- The zero-variance proposal is proportional to the integrand $\varphi_x(y) e^{-cg(y)}$ itself.
- Its first-order **Laplace / Gaussian** approximation gives the shift

$$\theta^* = -\sigma^2 c^{(k)} Dv^{(k)}(t, x).$$

- That is, samples are pushed **one preconditioned gradient step** along the descent direction of the running value iterate.

The previous Picard iterate hands the sampler its drift for free. Reachability and importance sampling become the *same* computation — the value gradient is the optimal sampling direction.

Effective Sample Size as a Self-Diagnostic

- Monitor the **effective sample size** of the self-normalized weights $\tilde{w}_i = w_\theta(y_i)e^{-cg(y_i)}$:

$$\text{ESS} = \frac{(\sum_i \tilde{w}_i)^2}{\sum_i \tilde{w}_i^2} \in [1, N].$$

- $\text{ESS} \approx N$: weights well-spread $\rightarrow$ estimate certified.
- $\text{ESS} \rightarrow 1$: **weight degeneracy** $\rightarrow$ flag, increase N , raise δ , or re-tilt.

ESS is a free, per-query health check: it tells you *when* to trust the estimate and *why* it failed when it does.

Part 4

Guarantees

Concentration · Contraction · Residual · Total Error · Certificate · Robustness.

▴ Four Error Sources, One Budget

The distance from the computed field $\hat{v}^{K,N,\delta}$ to the true inviscid viscosity solution v decomposes into four independently-controlled sources:

Source	Controlled by	Scaling
Iteration (Picard truncation at K)	More iterations	$C_1 \rho^K$
Monte-Carlo (finite N)	More samples	$C_2 N^{-1/2} \sqrt{\log(1/\delta_p)}$
Quasi-linearization residual $\mathcal{E}_{\text{ql}}$	Surrogate fidelity (not effort)	Floor
Viscosity (finite δ)	Smaller δ	$C_3 \sqrt{\delta}$

Three of four shrink with computational effort; the **residual is a property of the surrogate** and sets the floor. We now bound each.

▴ Finite-Sample Concentration (Theorem)

Fix phase (t, x) and frozen $c > 0$; assume $g_{\min} \leq g \leq g_{\max}$ on the sampling support. With $Z = e^{-cg(\zeta)}$, $\mu = \mathbb{E}[Z]$, $v_c = -\frac{1}{c} \log \mu$, $\alpha = e^{-cg_{\max}}$, $\beta = e^{-cg_{\min}}$, and $\hat{v}_{c,N} = -\frac{1}{c} \log \bar{Z}_N$:

$$\mathbb{P}(|\hat{v}_{c,N} - v_c| \geq \varepsilon) \leq 2 \exp\left(-\frac{2N\mu^2(1 - e^{-c\varepsilon})^2}{(\beta - \alpha)^2}\right).$$

- Bounded weights $\alpha \leq Z_i \leq \beta \Rightarrow$ Hoeffding.
- Rate is the standard $O(N^{-1/2})$, **independent of state dimension n** — the crux of the scalability claim.

Dimension enters the cost *per sample*, not the *number of samples* for a target accuracy. That is how we beat $O(M^n)$.

Explicit Sample Size, and Its Worst Case (Corollary + Remark)

- Since $\mu \geq \alpha$, it suffices to take

$$N \geq \frac{(\beta - \alpha)^2}{2\alpha^2(1 - e^{-c\varepsilon})^2} \log \frac{2}{\alpha}$$

to guarantee error $\geq \varepsilon$ with probability $\leq \alpha$.

- **Worst-case caveat:** with $c = 1/\delta$, $\beta/\alpha = e^{(g_{\max} - g_{\min})/\delta}$, so this bound grows like $e^{2(g_{\max} - g_{\min})/\delta}$ — **exponential in $1/\delta$** , while viscosity error is only $O(\sqrt{\delta})$.

The accuracy gain from shrinking δ and the sample cost pull **in opposite directions**. This Hoeffding bound is loose; the next slide tightens it, and experiments sit far below it.

Tightening: Bernstein & Jensen

- **Bernstein:** when $\text{Var}(Z) \ll (\beta - \alpha)^2/4$, a variance-aware tail bound gives strictly tighter concentration than Hoeffding.
- **Jensen:** the loose lower bound $\mu \geq \alpha = e^{-c g_{\max}}$ can be replaced by $\mu \geq e^{-c \mathbb{E}[g(\zeta)]}$, substantially reducing the required N .
- **Importance sampling** (Part 3) attacks the same variance directly.

The pessimistic $e^{1/\delta}$ is a worst-case artifact of Hoeffding + crude μ bound. In practice $N \in [14\text{k}, 20\text{k}]$ suffices at $\delta \in [0.08, 0.1]$.

The Iteration Operator (Contraction Setup)

- Fix evaluation states $x_1, \dots, x_M$ and time t ; let $\mathcal{G}$ be a stable gradient-reconstruction operator, $p_m(v) = (\mathcal{G}v)_m$.
- Coefficient-update map: $(\Gamma(v))_m = 2H(t, x_m, p_m(v))/(\delta|p_m(v)|^2)$.
- Frozen-coefficient heat-kernel map:

$$(\Phi(c))_m = -\frac{1}{c_m} \log \mathbb{E}_{\zeta \sim \mathcal{N}(x_m, \delta t I_n)} [e^{-c_m g(\zeta)}].$$

- Algorithm 1 is the iteration $v^{(k+1)} = \Lambda(v^{(k)})$, $\Lambda = \Phi \circ \Gamma$.

The whole method is a fixed-point iteration of a composed map on $\mathbb{R}^M$ with the sup-norm. Contraction $\Rightarrow$ convergence.

Assumptions for Contraction (Assumption A)

On a closed admissible set $\mathcal{A}$ (sup-norm), assume:

1. $|g| \leq G$;
2. non-degenerate gradient & bounded coefficient: $m_0 \leq |p_m| \leq P_*, c_{\min} \leq (\Gamma v)_m \leq c_{\max}$;
3. Hamiltonian Lipschitz in co-state: $|H| \leq H_*, |H(\cdot, p) - H(\cdot, q)| \leq L_H |p - q|$;
4. reconstruction Lipschitz: $\|\mathcal{G}v - \mathcal{G}w\|_\infty \leq L_D \|v - w\|_\infty$;
5. invariance + contraction constant

$$q = \frac{2G}{c_{\min}} \cdot \frac{2L_D}{\delta} \left(\frac{L_H}{m_0^2} + \frac{2H_*P_*}{m_0^4} \right) < 1.$$

Non-degeneracy (item 2) holds on the **tube band** $\{|v^\delta| \leq \eta\}$ where the gradient is bounded away from zero — where the barrier lives. Outside, use the regularized coefficient.

Contraction Convergence (Theorem)

Λ is a **contraction** on $\mathcal{A} \Rightarrow$ unique fixed point $v^\star$, and for any $v^{(0)} \in \mathcal{A}$:

$$\|v^{(k+1)} - v^\star\|_\infty \leq q \|v^{(k)} - v^\star\|_\infty \leq q^{k+1} \|v^{(0)} - v^\star\|_\infty,$$

with the **a posteriori** estimate

$$\|v^{(k)} - v^\star\|_\infty \leq \frac{q}{1 - q} \|v^{(k)} - v^{(k-1)}\|_\infty.$$

- Iteration count for tolerance ε : $K = O(\log(1/\varepsilon))$ — **geometric** convergence.

The residual you *watch* (successive-iterate change) bounds the error you *cannot* see (distance to $v^\star$).
That is the a posteriori estimate — a runtime stopping certificate.

Scope of the Contraction (Remark)

- The theorem is about the **frozen-coefficient numerical map**, not the original nonlinear PDE.
- The fixed point v^* solves the viscous HJ equation with H replaced by its **quadratic surrogate at v^*** .
- $v^* = v^\delta$ iff $H = \frac{1}{2}|p|^2$ (then $c \equiv 1/\delta$, $R_{\text{der}} \equiv 0$).
- For general H , the two differ by the **quasi-linearization residual** $\mathcal{E}_{\text{ql}} = \|v^* - v^\delta\|_\infty$.

We do **not** claim unconditional global convergence to the true HJ solution. We claim geometric convergence to a surrogate fixed point, plus a **bound on the gap** — next.

▢ The Quasi-Linearization Residual (Theorem, via Duhamel)

With converged coefficient $c^\star$, coefficient-variation bound L_c , value bound G_v :

$$\mathcal{E}_{\text{ql}} = \|v^\star - v^\delta\|_\infty \leq \frac{T}{c_{\min}} e^{2c_{\max}G_v} (\bar{R}_{\text{alg}} + \bar{R}_{\text{der}}),$$

- $\bar{R}_{\text{alg}} \leq c_{\max}(L_H + 2H_*P_*/m_0^2) \|Dv^\delta - Dv^\star\|_\infty$ — **contracts as the surrogate gradient tracks the true one.**
- $\bar{R}_{\text{der}} \leq L_c G_v(1 + \frac{\delta}{2}) + \frac{\delta}{2} L_c^2 G_v^2 + \delta(1 + c_{\max}G_v)L_c P_*$ — governed by **coefficient variation** L_c .

Proof idea: $\tilde{\omega} = e^{-c^\star v^\delta}$ solves the heat equation with a source ωB (Lemma); Duhamel + heat-semigroup L^∞ -contraction bounds $\|\tilde{\omega} - \omega^\star\|$; mean-value theorem for log converts back to v .

The bound **vanishes in the quadratic case** ($L_c = 0, Dv^\star = Dv^\delta$) and is largest where $H/|Dv|^2$ turns over sharply — the usable-part boundary.

Reading the Residual

- **Vanishes** when $H = \frac{1}{2}|p|^2$: recovers the closed-form Cole-Hopf.
- **Governed by L_c** , the variation rate of the converged coefficient — largest near the barrier where the Hamiltonian-to-gradient ratio shocks.
- **Algebraic part** $\propto \|Dv^\delta - Dv^\star\|$: the residual contracts as the Picard iteration refines the co-state.
- Certifies smallness where $c^\star$ is slowly varying; offers **no reprieve** where it is not.

This is why experimental errors concentrate at the zero-level-set boundary — the theory *predicts* the location of the worst error.

The Total Error Bound (Theorem) & Bias-Variance

Chaining all four sources by the triangle inequality:

$$\|\hat{v}^{K,N,\delta} - v\|_{\infty} \leq C_1 \rho^K + \frac{C_2}{\sqrt{N}} \sqrt{\log(1/\delta_p)} + \mathcal{E}_{\text{ql}} + C_3 \sqrt{\delta}.$$

- **Bias-variance tradeoff in δ :** viscosity bias $\downarrow$ as $\delta \downarrow$, but MC variance $\uparrow$.
- Balancing the two nontrivial δ -dependent terms yields the optimal

$$\delta \sim N^{-1/3} \Rightarrow \text{overall rate } O(N^{-1/6}),$$

slower than plain MC $O(N^{-1/2})$ but **dimension-robust and scalable**.

A single decomposition tells you how to spend budget: iterate to kill ρ^K , sample to kill $N^{-1/2}$, choose $\delta \sim N^{-1/3}$, and accept the residual floor.

The Conservative Safety Certificate (Corollary)

Let E be the total error budget. Declare state x :

- **SAFE** only if $\hat{v}^{K,N,\delta}(x) > E$;
- **UNSAFE** only if $\hat{v}^{K,N,\delta}(x) < -E$;
- **UNDETERMINED** otherwise ($|\hat{v}| \leq E$).

With probability $\geq 1 - \delta_p$: **no unsafe state is certified safe** and **no safe state is certified unsafe**; all classification error is confined to a declared band of width $2E$ about the boundary.

- The certificate **errs on refusal, never on admission**.
- This is the accuracy quotation: **worst-case sign-correctness with an explicit abstention band**, not an L^2 average that flatters the interior.

Robustness to Model Uncertainty

- The scheme is **stable under small perturbations** of the Hamiltonian and terminal cost (robustness theorems, appendix): perturb H by ΔH in sup-norm $\Rightarrow$ value perturbed by $\leq T \|\Delta H\|_\infty$ (Duhamel/comparison).
- Consequence: perfectly known dynamics and terminal costs are **not required** — real-world models with bounded error still yield certified-with-margin sets.
- The coefficient-clipping and regularization perturbations are of this priced form.

The method degrades **gracefully and quantifiably** under model error — a prerequisite for deployment on real robots with imperfect models.

Part 5

Experiments (from the Paper)

Rockets, Dubins, a 45D game, and 10^5 birds.



Experimental Setup and Statistics — 1/2

- **Hardware:** single CPU (Intel i7-14700K, 20 cores, 31 GiB RAM, Ubuntu 22.04); JAX on CPU backend.
— Consistent with the memory-frugality claim.
- **Replication:** Each experiment re-solved with **30 independent Monte-Carlo seeds**;
 - Evaluation points and the LevelSetPy reference held fixed — only sampler randomness varies.
Tables report mean $\pm$ 1 s.d.

Figures show one representative seed for clarity; tables report full 30-seed statistics. Multiplicity-controlled reporting throughout.



Experimental Setup and Statistics — 2/2

- **Significance:** Three families of **Holm-Bonferroni-corrected** tests at $\alpha = 0.05$:
 - (a) paired Wilcoxon of 30-seed-averaged MC field vs grid reference;
 - (b) one-sided t -test that per-seed L_{rel}^2 lies below the Crandall-Lions bound $\sqrt{\delta}$;
 - (c) cross-condition Mann-Whitney (Rockets vs Dubins; speed regimes).

Figures show one representative seed for clarity; tables report full 30-seed statistics. Multiplicity-controlled reporting throughout.

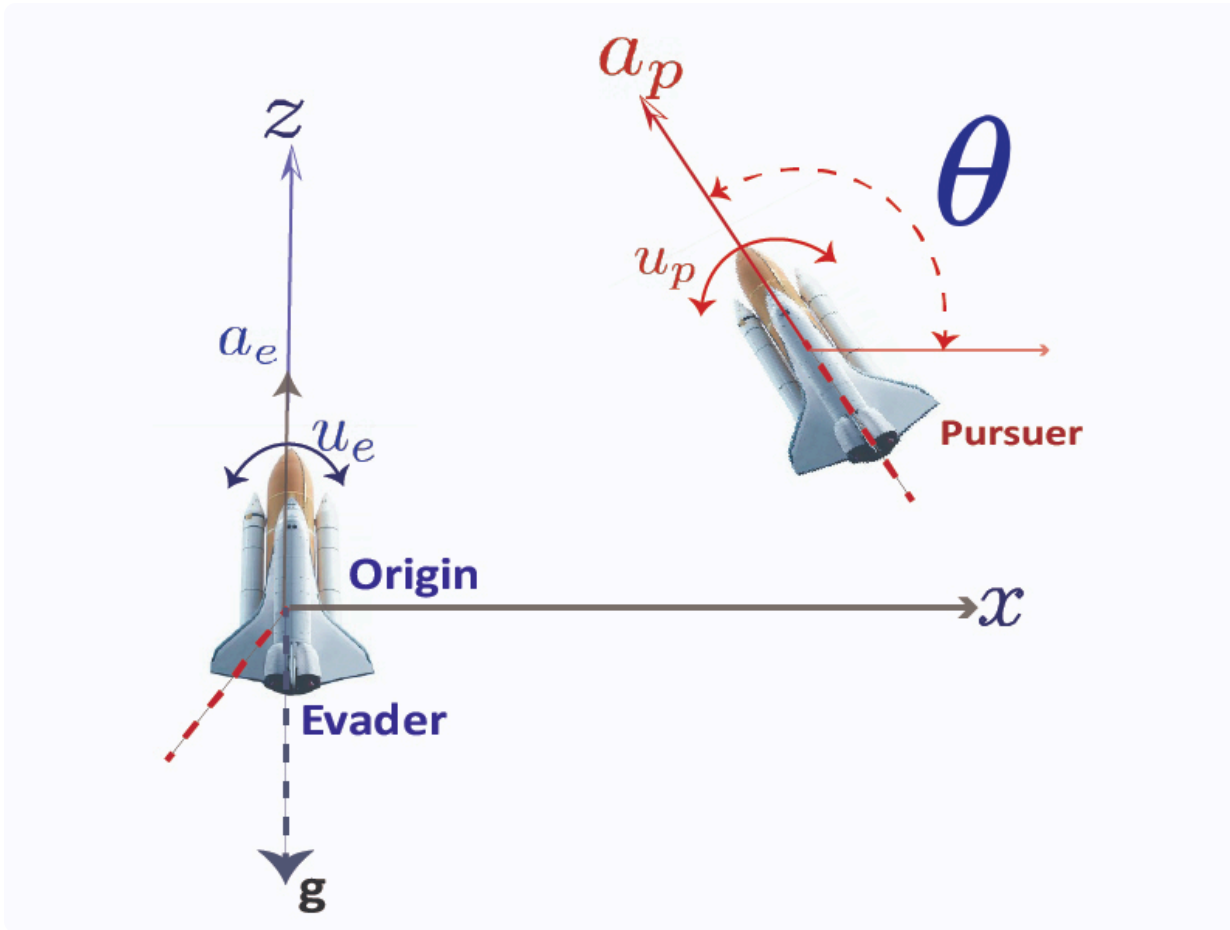
Benchmark 1: Two-Vehicle Pursuit-Evasion Games

- **Rockets launch game**¹: two identical rockets — pursuer P , evader E — on the x - z plane, thrusts (u_p, u_e) , relative orientation $\theta = u_p - u_e$. Capture when $\|PE\| < r$.
- **Dubins two-car game** (Merz): relative (x_1, x_2, θ) , symmetric turn-rate bound.
- Both are **3D relative-state** games; the BRT is a 3D tube, visualized as 2D (x, z) slices at fixed θ .
- Target: ℓ_2 -ball of capture radius $r = 1.5$; horizon $(0, 1]$; $\delta = 0.08$; $N = 14,000$ (rockets) / 20,000 (Dubins) samples per iteration.

These are the canonical HJ-Isaacs benchmarks — and the Dubins game is the pairwise MAPF conflict of Part 6.

¹ Dreyfus 1966

The Two-Rockets Problem (Schematic)



Relative geometry: evader E fixed at the origin (accel. a_e , control u_e , gravity g); pursuer P at relative orientation $\theta = u_p - u_e$ on the (x, z) plane. — HJ-Gauss (Molu et al., 2026).

Two-Rockets: Dynamics and Value Function

Relative state $x = (x, z, \theta)$ — evader with respect to pursuer, with $\theta = u_p - u_e$:

$$\dot{x} = a_p \cos \theta + u_e x, \quad \dot{z} = a_p \sin \theta + a_e + u_e x - g, \quad \dot{\theta} = u_p - u_e.$$

Writing $a \triangleq a_p = a_e$ and thrust bound $\bar{u} = \max(|u_p|, |u_e|)$, the Isaacs Hamiltonian is closed-form:

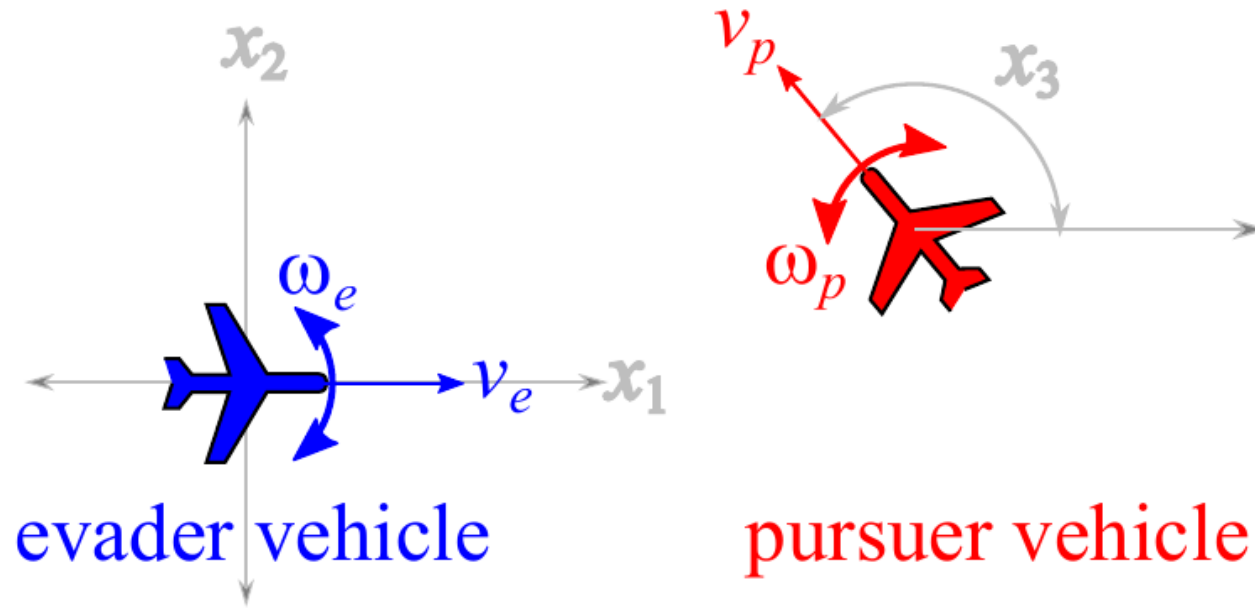
$$H(x, p) = -a p_1 \cos \theta - p_2 (g - a - a \sin \theta) - \bar{u} |p_1 x + p_3| + \bar{u} |p_2 x + p_3|.$$

The capture tube is the viscosity solution of the variational HJ-Isaacs equation:

$$v_t(t, x) + \min\{0, H(t; x, Dv)\} = 0, \quad v(0, x) = \|(x, z)\| - r.$$

- The two $|\cdot|$ terms **are** the optimal bang-bang thrusts, so no inner optimization is needed at run time.

The Dubins Air3D Problem (Schematic)



Air3D relative coordinates with the evader fixed at the origin: relative position (x_1, x_2) and relative heading ψ ; turn-rate-bounded pursuer vs evader. — LevelSetPy (Molu, ACM TOMS 2025 / IEEE CDC 2024).

Dubins Air3D: Dynamics and Value Function

Relative state $x = (x_1, x_2, x_3)$ — evader with respect to pursuer, x_3 the relative heading:

$$\dot{x}_1 = -v_e + v_p \cos x_3 + \omega_e x_2, \quad \dot{x}_2 = -v_p \sin x_3 - \omega_e x_1, \quad \dot{x}_3 = -\omega_p - \omega_e.$$

With a shared turn-rate bound ω (pursuer minimizing, evader maximizing), the Hamiltonian is again closed-form:

$$H(x, p) = p_1 (v_e - v_p \cos x_3) - p_2 v_p \sin x_3 - \omega |p_1 x_2 - p_2 x_1 - p_3| + \omega |p_3|.$$

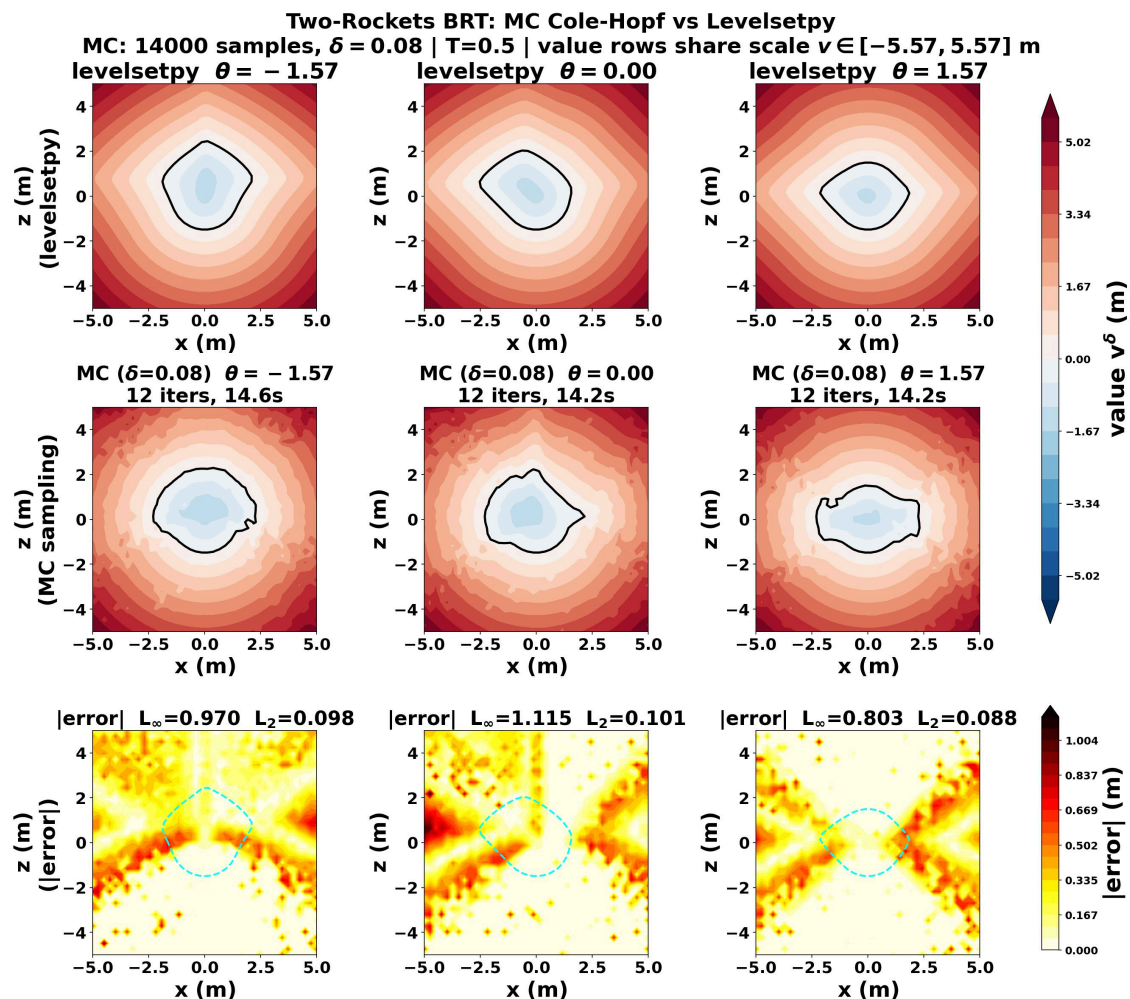
The conflict tube solves the same variational equation, with the capture cylinder as datum:

$$v_t(t, x) + \min\{0, H(t; x, Dv)\} = 0, \quad v(0, x) = \sqrt{x_1^2 + x_2^2} - r_c.$$

- The $\min\{0, \cdot\}$ freeze is what makes the zero sublevel set a **tube** rather than a set: once captured, always captured.



Rockets Pursuit-Evasion BRT

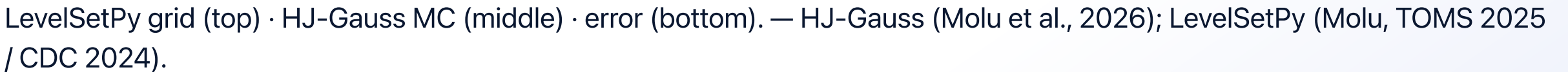


(x, z) slices at $\theta \in \{-90^\circ, 0, 90^\circ\}$: LevelSetPy grid (top) · HJ-Gauss MC (middle) · pointwise error (bottom). — HJ-Gauss (Molu et al., 2026).



Reading the Rockets Figure

- **Layout:** the three columns are heading slices $\theta \in \{-90^\circ, 0, 90^\circ\}$ of one 3D tube; the rows are **LevelSetPy grid** (top), **HJ-Gauss Monte-Carlo** (middle), and **pointwise error** (bottom).
- **What to look for:** the middle row recovers the top row's zero level set — the certified capture boundary — at every slice, with no grid ever stored.
- **Where the error lives:** the bottom row is near-zero across the interior and concentrates in a thin band **on the barrier**, exactly where the quasi-linearization residual bound says it should.
- **Physical read:** the $\theta = -90^\circ$ and $\theta = +90^\circ$ slices are **not** mirror images, because the gravity term $(g - a - a \sin \theta)$ leaves a $g - 2a$ drift at $+90^\circ$ against g at -90° .





Reading the Dubins Figure

- **Layout:** the same three-row comparison — grid, Monte-Carlo, error — for the Air3D game at the same three relative headings.
- **The signature shape:** the classic Merz notch appears where the turn-radius bound denies escape; a swept-disc test cannot produce that shape at all.
- **Agreement:** L_{rel}^2 runs 0.024 at $\theta = 0$ (smooth interior) to ≈ 0.13 at $\theta = \pm\pi/2$, while $L^\infty \approx 0.7$ -1.4 is set by the boundary where $|Dv^\delta|$ peaks.
- **Why it matters here:** this tube **is** the pairwise conflict predicate the MAPF shield queries, so the collision reduction rests on this agreement.



Benchmark 1: Quantitative Results

System	θ (rad)	L^∞	L^2_{rel}	MC time (s)	Iters
Rockets	$-\pi/2$	0.855 ± 0.047	0.098 ± 0.002	13.5	12
Rockets	0	1.034 ± 0.067	0.101 ± 0.001	13.5	12
Rockets	$\pi/2$	0.895 ± 0.109	0.090 ± 0.002	13.6	12
Dubins	$-\pi/2$	1.352 ± 0.002	0.131 ± 0.003	23.6	15
Dubins	0	0.701 ± 0.144	0.024 ± 0.001	23.7	15
Dubins	$\pi/2$	1.352 ± 0.002	0.132 ± 0.003	23.7	15

- Reference: LevelSetPy 45^3 grid interpolated to a 40×40 evaluation grid.
- Holm-Bonferroni one-sided t -test rejects $L^2_{\text{rel}} \geq \sqrt{\delta} = 0.283$ at $p_{\text{holm}} < 10^{-50}$ in **every** condition.

Sampling-plus-iteration error is statistically well inside the viscosity budget — not merely numerically.

Interpreting the Errors: Different Currencies

- The **Crandall-Lions** $O(\sqrt{\delta}) \approx 0.283$ bounds the *inviscid-vs-viscous* sup-distance; the table measures *MC-vs-grid* in L^2/L^∞ — **different currencies**, compared with care.
- **Averaging does not equal agreement**: a paired Wilcoxon of the 30-seed-averaged MC field vs the grid rejects equality at $p_{\text{holm}} < 10^{-8}$ everywhere — a systematic **quasi-linearization residual** remains (Theorem, Part 4).
- $\theta = 0$ Dubins reaches $L_{\text{rel}}^2 = 0.024$ (smooth interior); $\theta = \pm\pi/2$ and rockets sit ~ 0.09 - 0.13 (share of grid near the coefficient-turnover boundary).
- $L^\infty \sim 0.7$ - 1.4 driven by the zero-level-set boundary where $|Dv^\delta|$ is maximal — where the residual L_c term predicts.

What we report is the **Corollary (conservative certificate)**: sign-correctness outside an abstention band; L^2 flatters the interior, L^∞ is dominated by the declared-undetermined band.



A Physical Check: The Gravity Asymmetry

- Rockets are asymmetric between $\theta = -90^\circ$ and $\theta = +90^\circ$: the gravity term ($g - a - a \sin \theta$) gives a $g - 2a$ drift at $+90^\circ$ but g at -90° .
- Holm-Bonferroni test over 30 seeds detects this asymmetry for **Rockets** ($p_{\text{holm}} < 10^{-9}$) but **not** for the gravity-free **Dubins** vehicle ($p_{\text{holm}} = 0.53$).

The method resolves a **genuine physical effect**, and the statistics correctly find it present where physics says it should be and absent where it should not. Validation, not just fitting.



Benchmark 2: A 45-Dimensional Pursuit-Evasion Game

- **15 rockets** (14 pursuers, 1 evader), state $x \in \mathbb{R}^{45}$: each agent (x_i, y_i, θ_i) , control $u_i \in [-1, 1]$.
- Dynamics $\dot{x}_i = a_i \cos \theta_i$, $\dot{y}_i = a_i \sin \theta_i$, $\dot{\theta}_i = u_i$; target $\phi(x) = \min_{i \leq 14} \|x_i^{\text{pos}} - x_{15}^{\text{pos}}\| - r_{\text{capture}}$.
- Three regimes: **evader faster** (2.0 vs 1.0), **balanced** (1.0/1.0), **pursuers faster** (1.0 vs 2.0).
- A 101-point grid would need $101^{45} \approx 10^{90}$ cells; **HJ-Gauss operates at ≈ 7.2 MB per iteration**.

This is the flagship: a reachability computation **grid solvers cannot even represent**, done on a single CPU in seconds.



Benchmark 2: 45D Results

Case	a_{evader}	a_{pursuers}	Iters	Residual $\varepsilon(k)$	Wall-clock (s)
Evader faster	2.0	1.0	15	0.0002 ± 0.0005	12.7
Equal speed	1.0	1.0	15	0.0003 ± 0.0005	12.6
Pursuers faster	1.0	2.0	15	0.0006 ± 0.0013	12.6

- Memory constant at **7.2 MB/iteration**; residuals below 10^{-2} (indeed $\sim 10^{-4}$) → **near-zero Picard residual floor**.
- Min-max Hamiltonian evaluates in **closed form** (box-constrained control $\Rightarrow$ sign structure), so each iteration is $O(N \cdot n)$ arithmetic and memory.
- Pairwise Holm-Bonferroni Mann-Whitney: **no significant residual difference** between speed regimes ($p_{\text{holm}} \geq 0.93$) → floor set by (N, δ) , not game parameters.

At $n = 45$ there is **no grid reference**; we report *measurable* quantities — iteration stability, memory, wall-clock — i.e. **scalability, not certified accuracy**.

Why the 45D Residual Floor Is *Lower* Than 3D

- 3D benchmarks floor at 0.02-0.06; the 45D game floors at 0.0002-0.0006 — **two orders of magnitude lower**.
- Reason: the residual floor tracks **proximity to the coefficient-turnover region** (the shock-prone barrier), not dimension per se.
- 45D evaluation states are drawn uniformly over $[-100, 100]$ per coordinate — with **small probability of landing near the boundary** — so the average residual is small.

A subtle but important message: **error concentrates geometrically (at the barrier), not dimensionally**. High dimension is not inherently high error.



Cost Accounting: Time vs Memory

- 3D isosurface over 15,625 points: **129.4 s** (rockets, single CPU) vs **1.3 s** for the dense $45^3 = 91,125$ -point LevelSetPy grid.
- Per-iteration memory: ≈ 0.6 MB (MC, $N = 14\text{k}$, $n = 3$) vs ≈ 1.5 MB (grid value+gradient).
- **In low dimensions ($n \leq 4$) grids win** — smaller constant, no sampling variance.
- The Monte-Carlo overhead **becomes favorable where grid storage becomes prohibitive** ($n \geq 5$).

The two paradigms are **complementary, not competing**. Use grids where they fit in memory; use HJ-Gauss where they cannot.



Benchmark 3: Safety at Population Scale (10^5 Starlings)

- Certify **100,000** European starlings (*Sturnus vulgaris*) as 4D aerial Dubins vehicles under predator attack.
- **Structure (explicit)**: the murmuration is partitioned into **flocks**; each flock has its **own** value function; the population safe set is the aggregation of per-flock zero sublevel sets — **many coupled low-D games, not one joint high-D PDE**.
- Because the solve is grid-free, the value-function solve is **independent of bird count**; the population enters only through **parallel per-bird certification** — the operational content of $O(N \cdot n)$ at 10^5 agents.

Two curses dodged at once: **dimensionality** (grid-free solve) and **agent cardinality** (per-flock decoupling + parallel evaluation).

The Repertoire Is Field-Documented

- Real starling responses to predation: **cordon**, **tube**, **funnel**, and **split**.
- Each one appears in our certificates as a **topological signature** of the safe set, and none of them is a hand-coded rule.
- The certificate reproduces a documented repertoire from **dynamics and the game alone** — which is the reason to trust it on a floor, where the repertoire is congestion, yielding, and deadlock instead.

Fragmentation event, $n_c : 1 \rightarrow 2$. Field murmuration imagery as reproduced in HJ-Gauss (Molu et al., 2026).





Behaviour 1: Dense Ascent From a Roost

- Thousands of birds leave a roost at once, and the flock's **local density spikes** while every bird is still turn-rate and climb-rate limited.
- **The biological question:** can a bird in the interior still retain an escape maneuver when its near-neighbour spacing collapses?
- **What we certify:** whether the interior states stay outside the capture set for the whole window — the dense-packing analogue of a congested aisle.

Series photo credits: Reuters/Amir Cohen · AP
Photo/Oded Balilty · Menahem Kahana/AFP/Getty Images
· Courtesy of The Gathering Site — reproduced in
LevelSetPy (Molu, ACM TOMS 2025) and HJ-Gauss
(Molu et al., 2026).

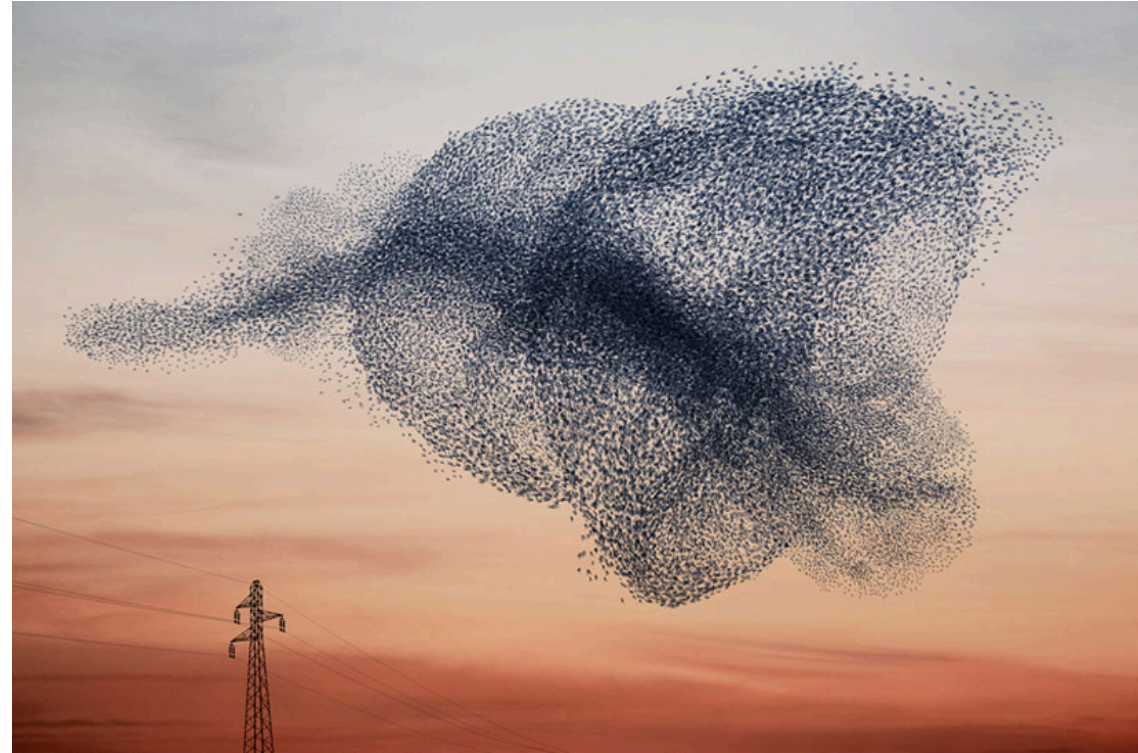




Behaviour 2: The Concentric Cordon

- Under sustained pressure the flock arranges itself into a **concentric, tube-like shell** around an interior core.
- **The biological question:** is the core genuinely protected, or does it merely look protected from outside?
- **What we certify:** an **annular** safe set, first Betti number $\beta_1 = 1$ — a hole the predator cannot certify entry into. This is the cordon marker.

Series photo credits as on the previous slide.





Behaviour 3: Funnel and Split

- A pressed flock **necks into a funnel** and can divide, leaving a subgroup separated from the main body.
- **The biological question:** which birds end up isolated, and when does isolation become unavoidable?
- **What we certify:** a rise in the connected-component count n_{comp} — fragmentation, the marker that says a subgroup has been cut off rather than merely crowded.

Series photo credits as on the first slide of this series.





The Bird Model: 4D Aerial Dubins

Each bird carries state $(x_1, x_2, x_3, \theta) \in \mathbb{R}^2 \times \mathbb{R} \times \mathbb{S}^1$ — planar position, altitude, heading:

$$\dot{x}_1 = v \cos \theta, \quad \dot{x}_2 = v \sin \theta, \quad \dot{x}_3 = u_z, \quad \dot{\theta} = \langle \omega \rangle_r.$$

- The **coupling lives in the heading**: each bird turns at the neighbour-averaged rate

$$\langle \omega \rangle_r = \frac{1}{1 + n_i} \left(\omega_i + \sum_{j \in \mathcal{N}_i} \omega_j \right), \quad n_i = |\mathcal{N}_i|.$$

- Alignment is therefore a property of the **dynamics**, not a penalty added to a cost.
- Bounded controls: $|u_z| \leq \gamma_{\max}$ for climb rate, $|\omega| \leq \bar{\omega}$ for turn rate.

The Attacked-Flock Game

Relative state of an attacked bird with respect to its predator:

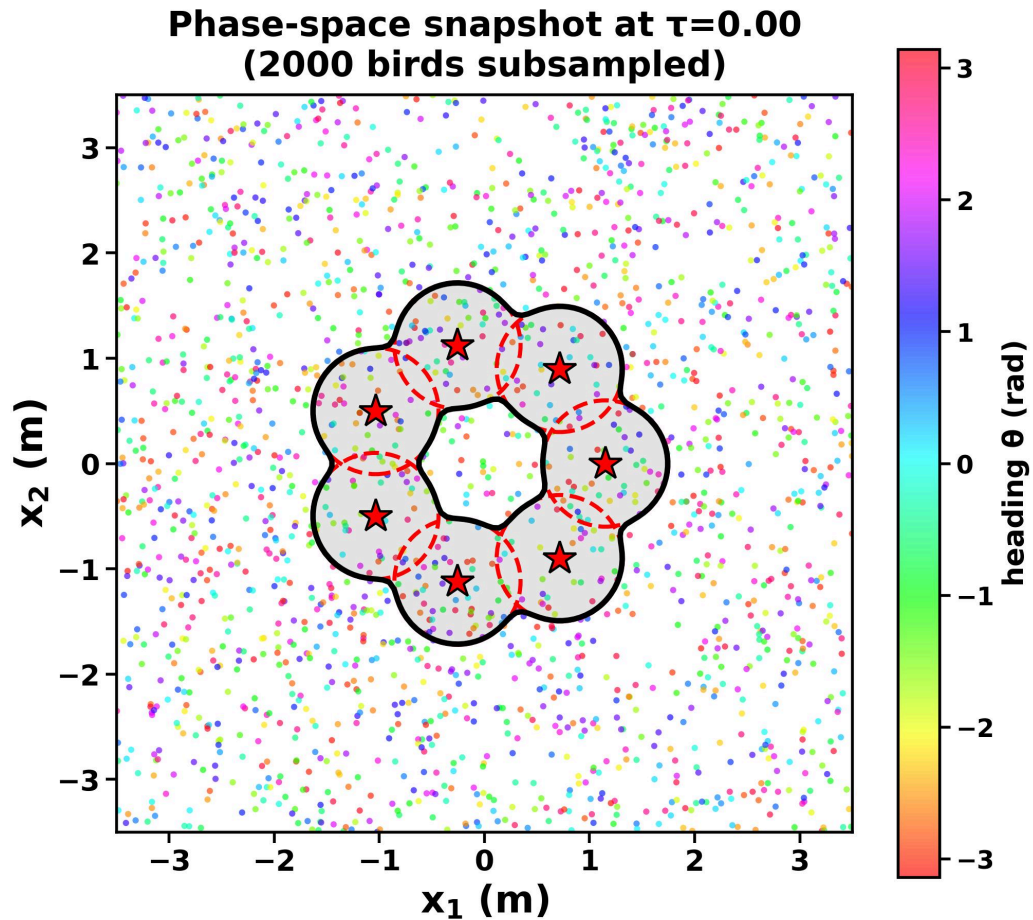
$$\begin{aligned}\dot{x}_1 &= -v_p + v_e \cos \theta + \langle \omega_e \rangle_r x_2, & \dot{x}_2 &= v_p \sin \theta - \langle \omega_e \rangle_r x_1, \\ \dot{x}_3 &= u_z^e - u_z^p, & \dot{\theta} &= \omega_p - \langle \omega_e \rangle_r.\end{aligned}$$

Datum is the capture cylinder $g(x) = \sqrt{x_1^2 + x_2^2} - r_c$, and each flock solves its **own** variational equation:

$$v_t + \min\{0, H_{\text{att}}(x, Dv)\} = 0, \quad H_{\text{att}} = \min_{u_p} \max_{u_e} Dv^\top f(x, u_p, u_e).$$

- Run shown here: **99,996 starlings** in six flocks, **7 predators** on a ring of radius 1.15, $r_c = 0.6$, $\delta = 0.18$, $N = 1600$ samples, ≤ 7 Picard iterations, 128×128 window, 18 backward-time steps.
- Whole study, value solves plus certifying every bird: **429 s on eight CPU cores**, no GPU. **87.7%** of birds end certified outside the capture set.

The Certified Population Snapshot



2,000 of 99,996 birds (colored by heading) with 7 flock centers (stars) and their capture discs (dashed). Black contour is the certified safe-set boundary at $\tau = 0$: an **annular cordon** around a protected core. — HJ-Gauss (Molu et al., 2026).



Reading the Population Snapshot

- **Coloured dots:** 2,000 of the 99,996 birds, subsampled for legibility and coloured by heading θ . The spread of colour is the point — the population is a **heading distribution**, not a rigid formation.
- **Stars and dashed circles:** the seven predators and their capture radii $r_c = 0.6$.
- **Black curve:** the $v = 0$ boundary at $\tau = 0$. The shaded zero-sublevel set $\{v \leq 0\}$ is the flock's **certified safe set**.
- **The hole is the message:** an annular safe set ($\beta_1 = 1$) is a **defensive cordon** — a protected core the predators cannot certify entry into.



What the Snapshot Certifies

- Each bird's status is one **membership query** against the cached value field: sign of v at its own 4D state.
- So the 99,996 birds cost 99,996 **independent** $O(1)$ lookups, trivially parallel — while the solve itself never represented 10^5 agents.
- That is the operational meaning of $O(N \cdot n)$: the certificate's cost scales with the **sample budget and state dimension**, not with fleet size.

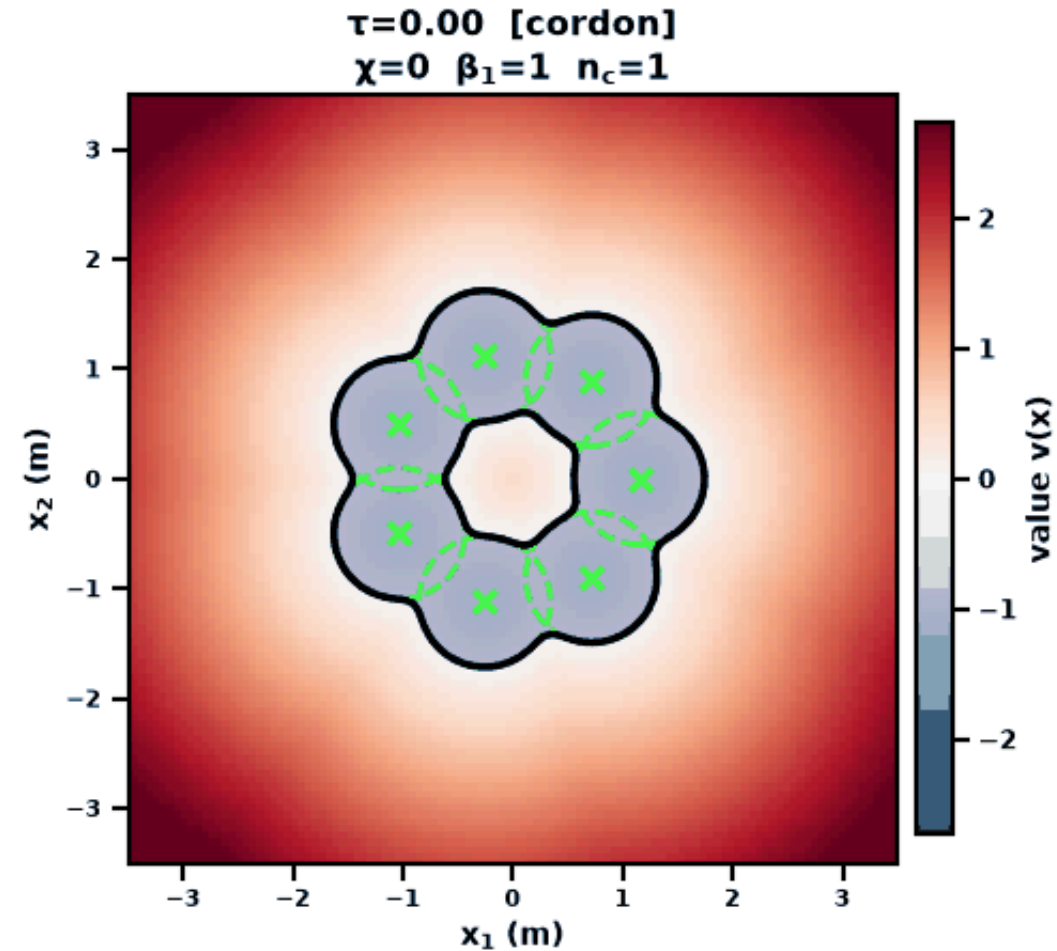
The warehouse analogue: certify a thousand drive units against a per-zone value field, with no joint state space anywhere in the loop.



Watch the Safe Set Change Shape

- Backward time τ runs outward from the capture set: the certified safe set of the attacked flock **grows and changes topology**.
- Green crosses and dashed circles are the seven attacking predators with their capture discs.
- Blue interior = certified safe; the black curve is the $v = 0$ boundary.

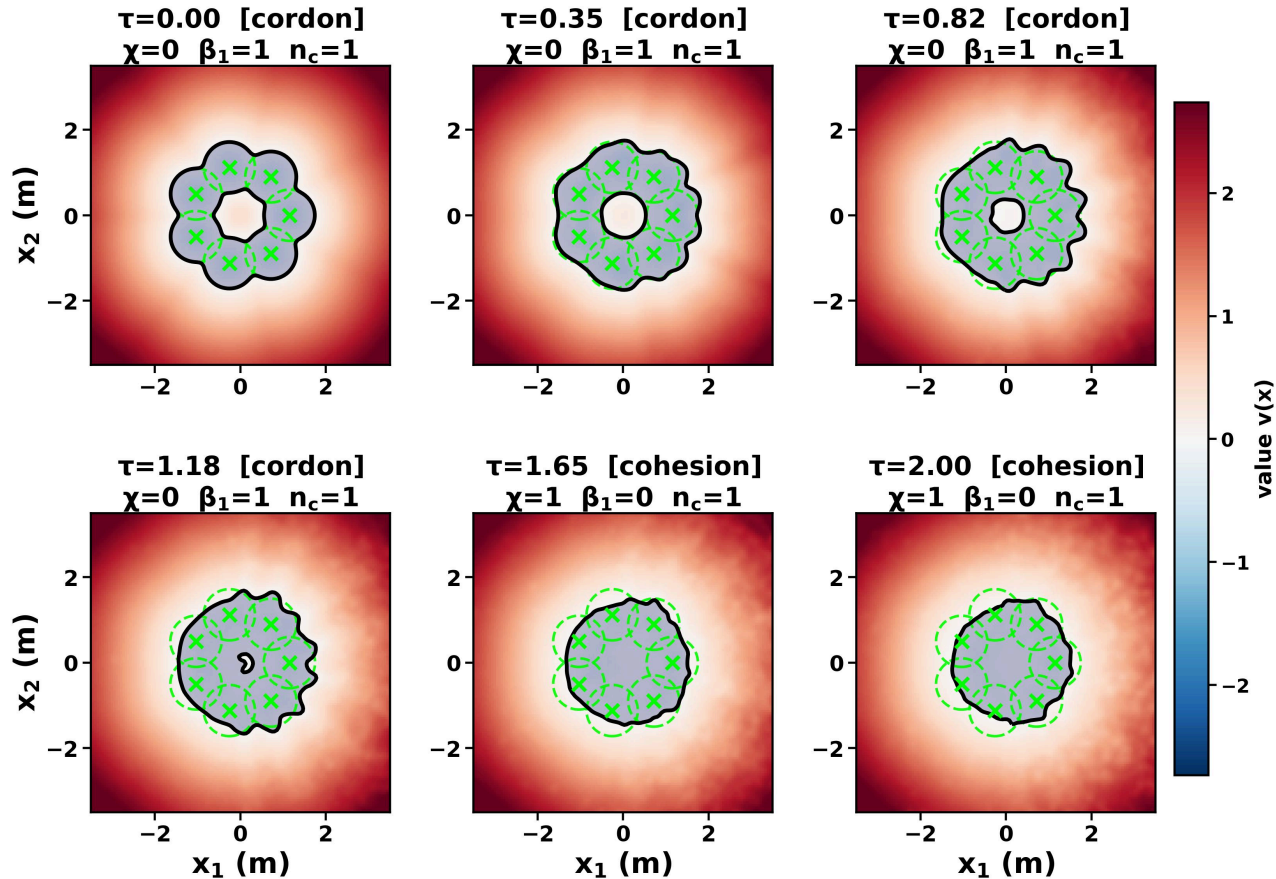
A safety certificate for 10^5 agents, as one evolving surface.





Cordon, Then Collapse

BRT zero-level-set evolution (backward time τ)



Same solve, six instants. A hole persists in the safe set ($\chi = 0$, $\beta_1 = 1$, **cordon**) through $\tau = 1.18$; by $\tau = 1.29$ it has closed ($\chi = 1$, $\beta_1 = 0$, **cohesion**). — HJ-Gauss (Molu et al., 2026).

Topology as a Safety Instrument

The certificates recover the field-documented collective repertoire as **topological events** of the reachable set, compressed to three integers per time step $(\chi, \beta_1, n_{\text{comp}})$:

Behavior	Topological signature	Marker
Vacuole nucleation (predator penetrates)	Drop in Euler characteristic χ	Threshold crossing in χ
Defensive cordon (protected core)	Annular safe set, first Betti number $\beta_1 = 1$	β_1 transition
Flock fragmentation	Rising connected-component count n_{comp}	Component-count jump

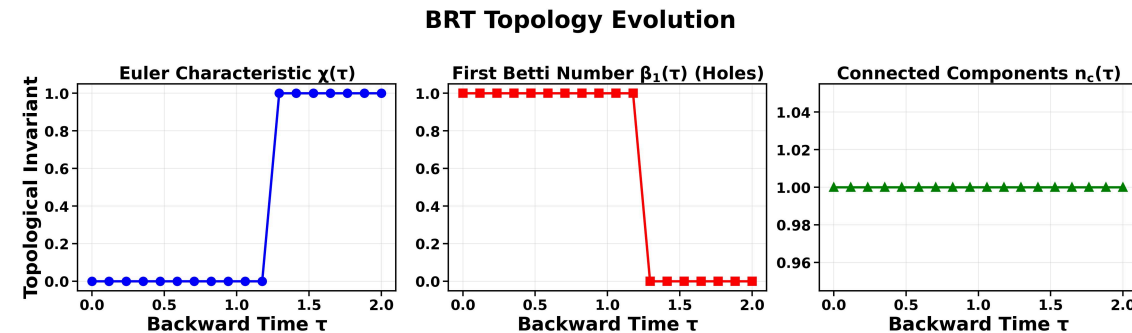
The triple $(\chi, \beta_1, n_{\text{comp}})$ tells an operator not merely **that** safety is being lost but **how** — nucleation vs cordon vs split. Safety posture as a 3-integer summary.



The Three Integers, Over Time

- χ steps $0 \rightarrow 1$ while β_1 drops $1 \rightarrow 0$ at the **same** $\tau = 1.29$: one event, two independent witnesses.
- n_c holds at 1 throughout, so this flock never fragments.
- **Vacuole nucleation** is the reverse event: a predator penetration attaches a 1-handle and drops χ by one.

Three integers per tick say whether safety is being lost, and **how**.



Part 6

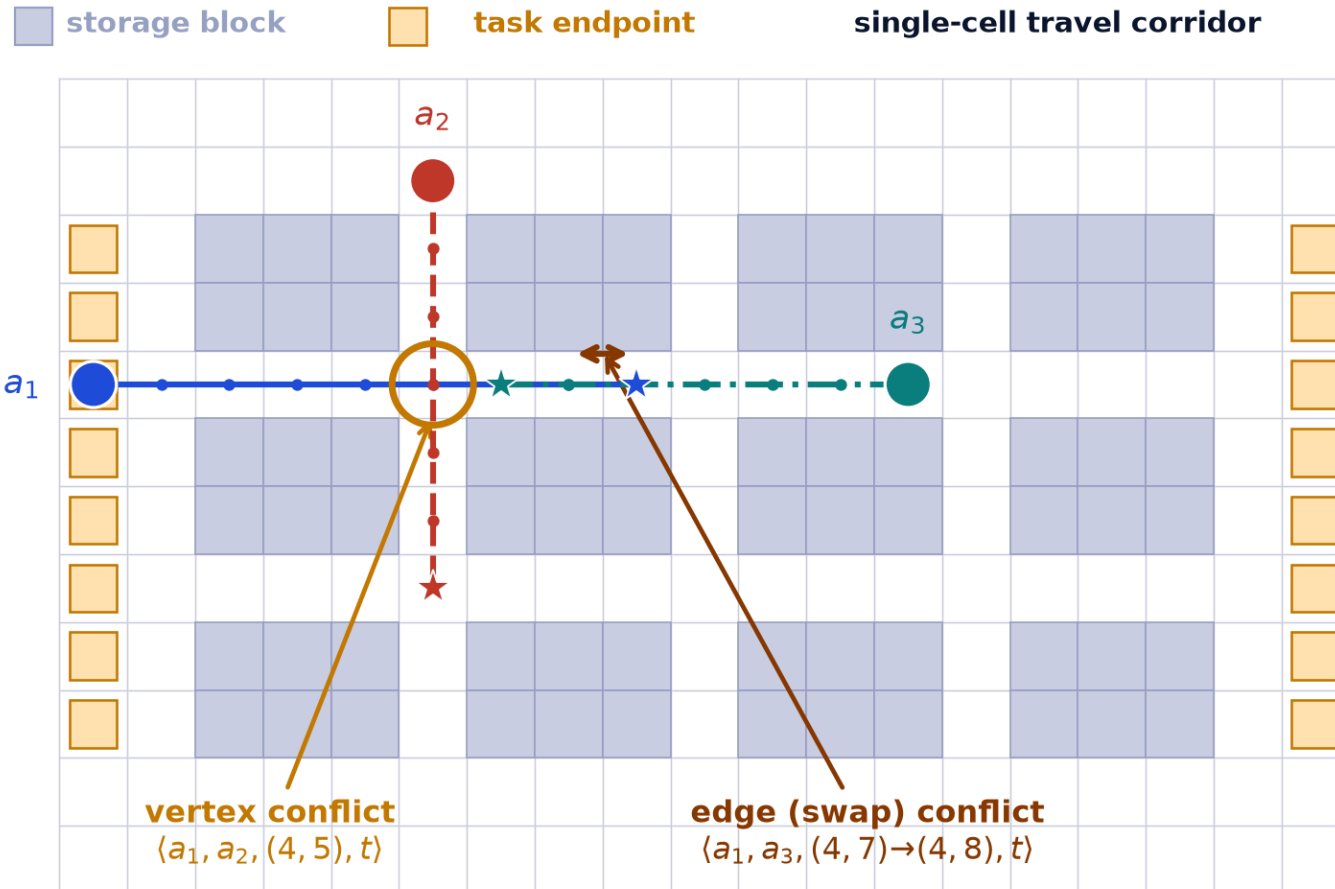
Application: Multi-Agent Path Finding

The reason this talk is for the autonomous mobility team.



The Floor, as MAPF Sees It

Discrete MAPF: the grid-partitioned floor that CBS / ECBS / RHCR search over



Storage blocks separated by single-cell travel corridors, task endpoints on the border columns — the RHCR fulfillment-warehouse map family (Li et al., AAAI 2021). CBS/ECBS resolve the two classical conflict types marked above.



What Counts as a Conflict

- **Vertex conflict** $\langle a_i, a_j, x, t \rangle$: two agents occupy the same cell at the same timestep.
- **Edge (swap) conflict** $\langle a_i, a_j, x \rightarrow y, t \rangle$: two agents traverse the same edge in opposition — the head-on case in a one-cell corridor.
- **CBS** branches on a conflict, adds a constraint to one agent, and re-plans that agent's shortest path; **ECBS** bounds suboptimality; **RHCR** applies this inside a rolling window.
- Every one of these predicates is a statement about **cells and timesteps**.

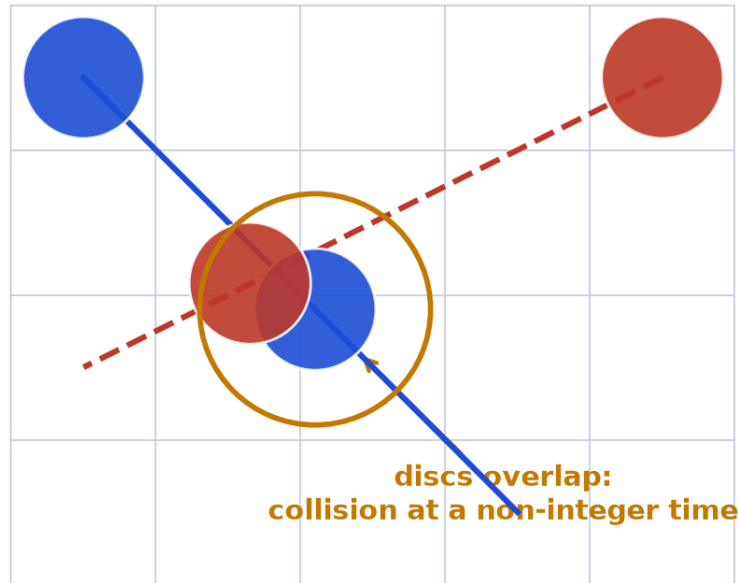
Nothing in this vocabulary mentions turn radius, actuation lag, or localization error.



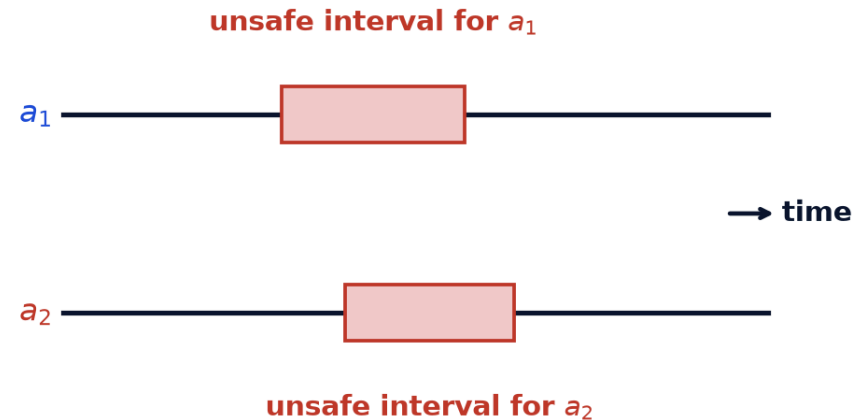
CCBS Fixes Time, Not Dynamics

CCBS: geometric agents in continuous time

straight-line moves, arbitrary durations, 2^k -neighbourhood



Conflicts as unsafe time intervals



Constraints become time intervals, not integer timesteps.
Still absent: turn radius, actuation lag, localization noise.

CCBS (Andreychuk et al., IJCAI 2019) replaces integer timesteps with continuous time: agents are discs moving along straight edges with arbitrary durations, and conflicts become unsafe time intervals. Turn radius, actuation lag, and disturbance remain outside the model.



The Gap in Classical MAPF

- **Multi-Agent Path Finding (MAPF)**: move a team of agents to goals without collision. Solvers: **CBS/ECBS** (conflict-based search), **PIBT**, **LaCAM/LaCAM***, and lifelong **RHCR**.
- All operate on a **discrete grid with simplified kinematics** (unit moves / rotate-in-place) and **no disturbance model**.
- A "conflict" is a shared vertex/edge (or swept-disc overlap) — **dynamics-blind**.
- Consequence: realized collisions on a real floor (turn-radius limits, localization noise, actuation lag) are **out of scope**.

The discrete plan is provably conflict-free *on the graph* — yet the continuous robot can still collide. That is the gap HJ reachability fills.



The Scaling Anchor: RHCR¹

- **Rolling-Horizon Collision Resolution:** decompose lifelong MAPF into a sequence of **Windowed MAPF** instances; resolve collisions only within a bounded time window w , ignore beyond.
- Scales to **1,000 agents** on warehouse maps; co-authored by **J. Durham (Amazon Robotics)**.
- Current frontier: **LaCAM*** (10,000+ agents, near-optimal, anytime), **PIBT** (thousands in ms), learning-based lifelong (imitation, guidance-graph, MAPF-GPT).
- **All still discrete-grid, deterministic.**

HJ-Gauss does not compete with LaCAM's combinatorial search; it **certifies** the plan under continuous dynamics — a different, complementary axis.

¹ Li et al., AAAI 2021

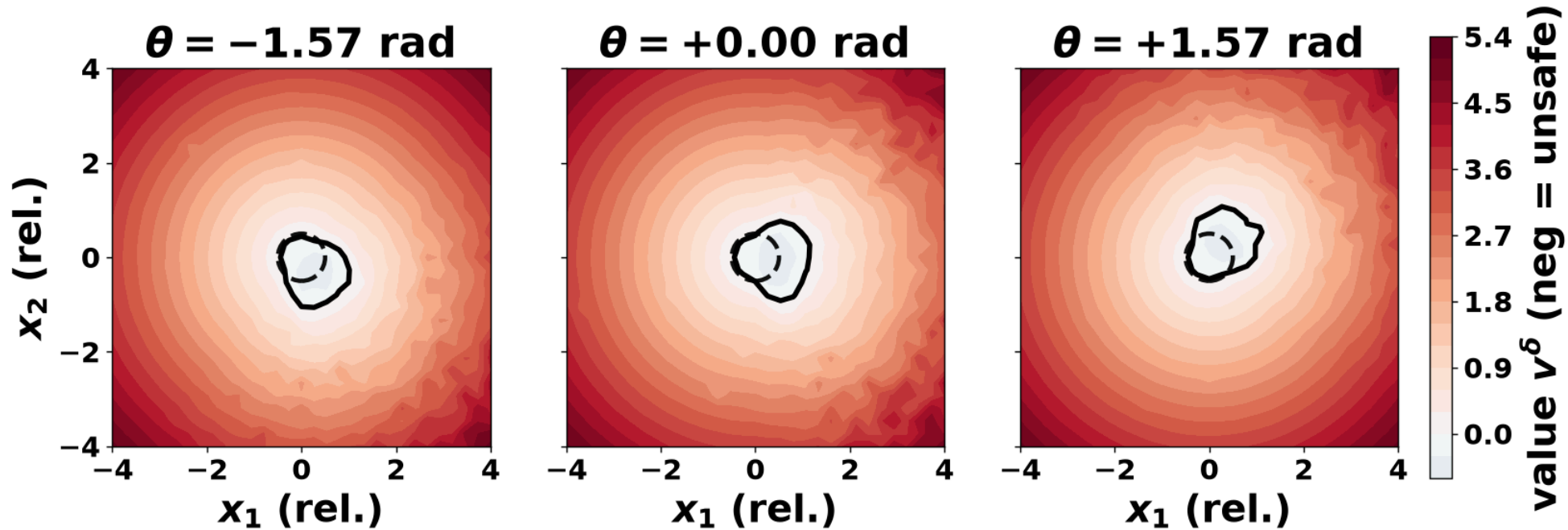
The Mapping: HJ-Gauss Constructs → MAPF Roles

HJ-Gauss construct	MAPF role
Windowed BRT over $(0, T]$, set $T = w$	Continuous, disturbance-robust replacement for the RHCR rolling window (innovation #1).
Pairwise Dubins BRT membership	Dynamics-aware conflict predicate replacing the geometric/disc check in CBS/CCBS (innovation #2).
Relative-frame tube, computed once	Reused across all pairs & timesteps $\Rightarrow$ certificate cost $O(1)$ in agents .
Value gradient $Dv \propto \nabla \log p$	A score : certified guidance term for diffusion planners (innovation #3).

The Dubins pursuit-evasion BRT from Part 5 **is** the canonical pairwise MAPF conflict, computed rigorously.

The Dynamics-Aware Conflict Set (Pairwise Dubins BRT)

Pairwise Dubins BRT (black = inevitable-collision boundary) vs naive disc (dashed)



Black = certified inevitable-collision boundary; dashed = naive disc. The BRT bulges beyond the disc in a **heading-dependent** way — what a dynamics-blind check cannot see.

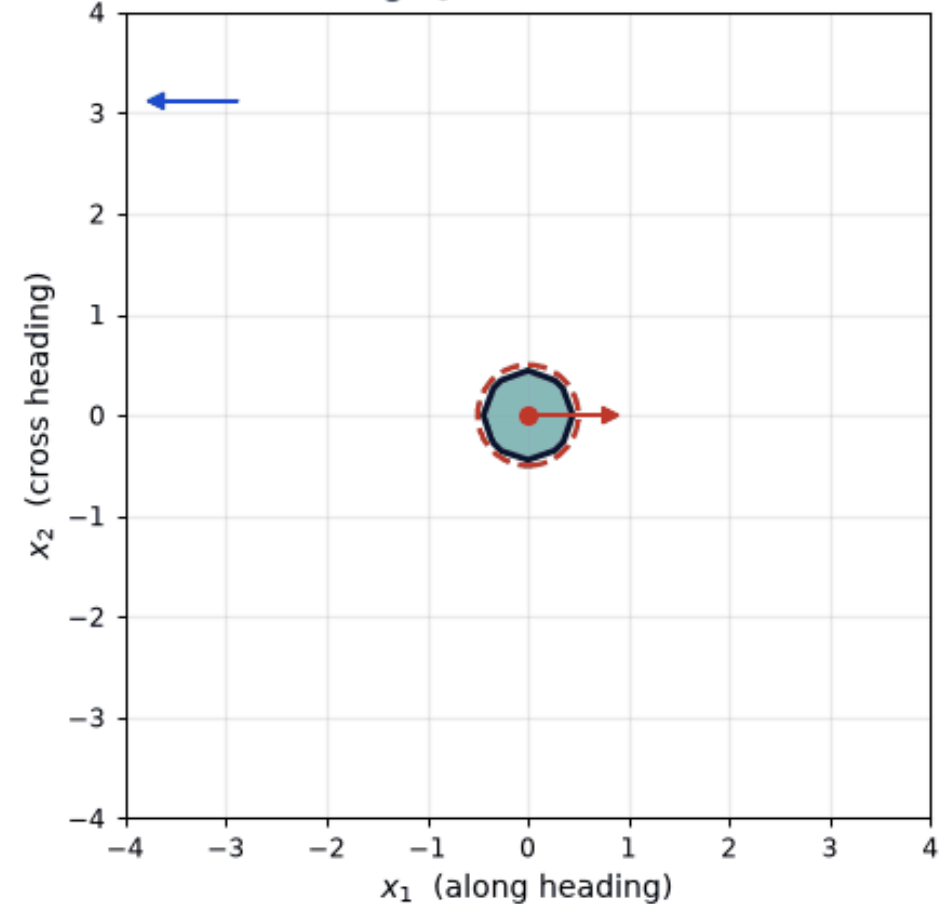
⚙ What the Certified Set Looks Like

- Sweeping the relative heading x_3 makes the certified set **rotate and deform**.
- A near head-on geometry grows a long lobe; a near-parallel one stays compact.
- The dashed red circle is the dynamics-blind disc, **fixed for every heading**.

This is the object a swept-disc test approximates with a circle.

Windowed pairwise conflict set, $w = 0.6$ s

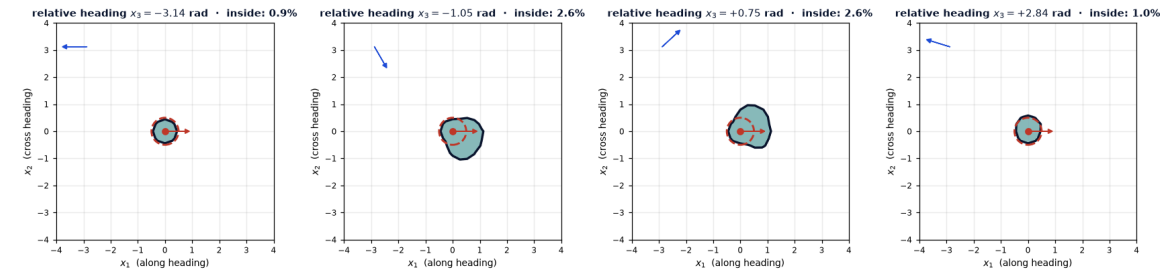
relative heading $x_3 = -3.14$ rad • inside: 0.9%



teal = certified inevitable collision • dashed red = dynamics-blind disc

⚙ Reading the Heading Sweep

- Only about **2%** of the relative-state box is certified inevitable-collision at $w = 0.6$ s.
- So the gate is **selective**, not blanket-conservative: it fires on geometry that actually traps the pair.
- Where the tube exceeds the disc, a geometric planner is blind; where it falls inside, the disc is needlessly cautious.



Falsifiable Hypotheses (Stated to Be Proven Wrong)

Naming: **HJ** = Hamilton-Jacobi (the method); **HB- n** = hypothesis n .

ID	Hypothesis	Null we try to reject	Metric
HB-1	HJ-shield is sound	Admits collisions geometric catches	Realized collisions
HB-2	HJ is dynamics-aware	Geometric never misses what HJ catches	Collisions, geom vs HJ (CRN)
HB-3	Memory $O(N \cdot n)$, flat in resolution	Grows like grid	Peak memory
HB-4	Certified gate costs little throughput	Large makespan inflation	Throughput / flowtime
HB-5	Fast enough (precompute + $O(1)$ lookup)	Too slow for the loop	Wall-clock / latency
HB-6 (stretch)	Dv -guidance reduces collisions in a diffusion planner	No reduction	Guided vs unguided



The AMFS Pipeline: World and Planner

- **World:** an RHCR-style structured warehouse grid — storage blocks, single-cell corridors, border endpoints — with **lifelong** task reassignment.
- **Planner:** windowed prioritized planning (space-time A^*), the RHCR rolling horizon, **shared by both policies**.
- **Executor:** a continuous unicycle/Dubins rollout with bounded turn rate and bounded disturbance, where realized collisions are detected in continuous space.

The planner is held fixed on purpose. Only the safety layer varies.



The AMFS Pipeline: The Two Policies

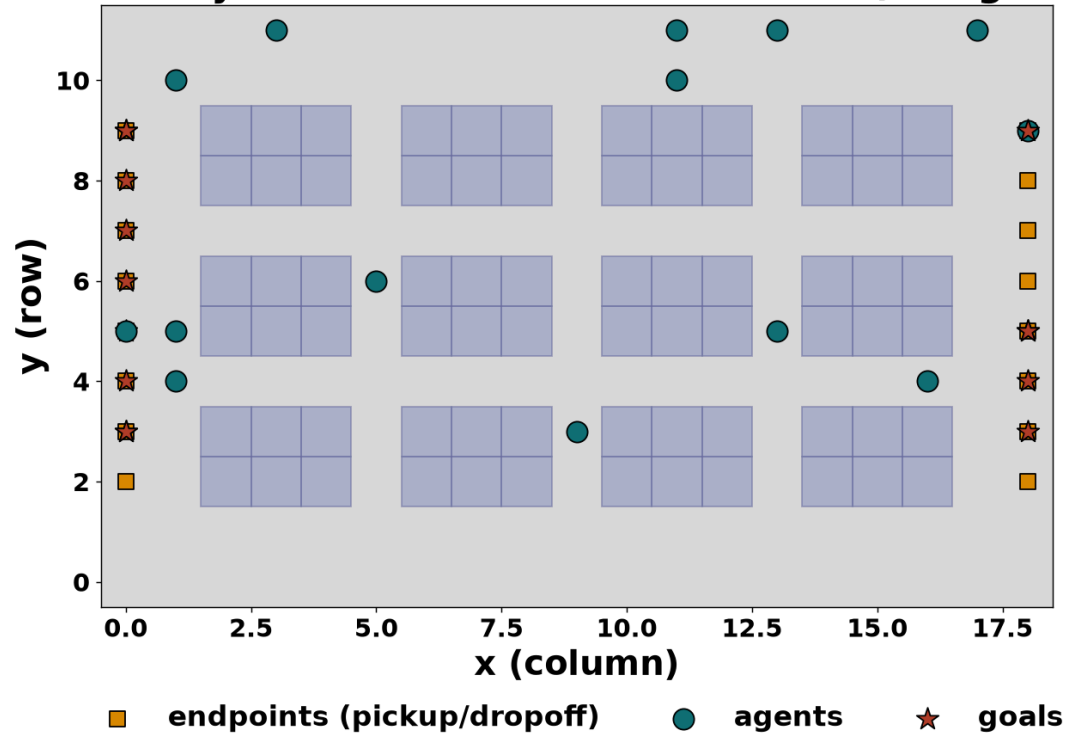
- **Geometric** — the classical dynamics-blind vertex/edge conflict test, unshielded.
- **HJ** — the same planner **plus** the windowed-BRT runtime shield.
- Both run under **Common Random Numbers**: identical spawns, tasks, priorities, and disturbance realizations for a given seed.
- **BRT precompute**: the JAX HJ sampler (Algorithm B) on the Dubins-relative Hamiltonian, cached once; the online loop is numpy-only $O(1)$ lookups.

One clean contrast: the only difference between the two arms is whether the certificate is actuated.



The Simulated Floor

RHCR-style structured warehouse floor (14 agents)

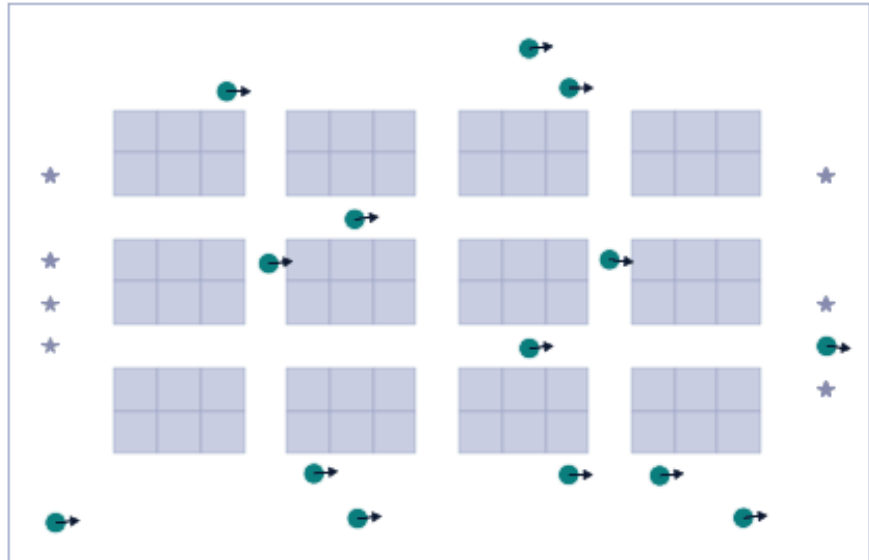


RHCR-style structured floor: storage blocks (obstacles), border endpoints (pickup/dropoff), 14 lifelong agents (circles) heading to goals (stars).

▶ Watch It Run: Same Seed, Same Noise, One Difference

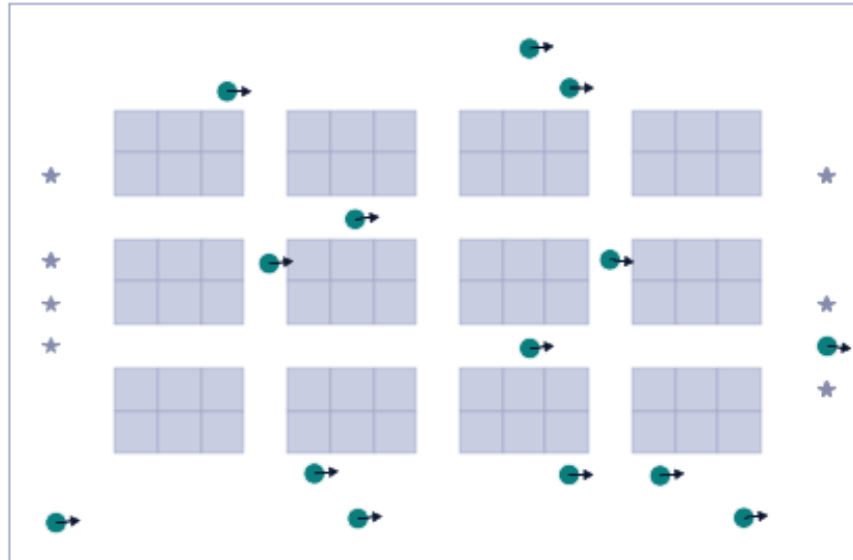
AMFS structured floor · 14 lifelong agents · same seed, same disturbance stream (t = 0.0 s)

Geometric conflict check (dynamics-blind)



realized collisions: 0

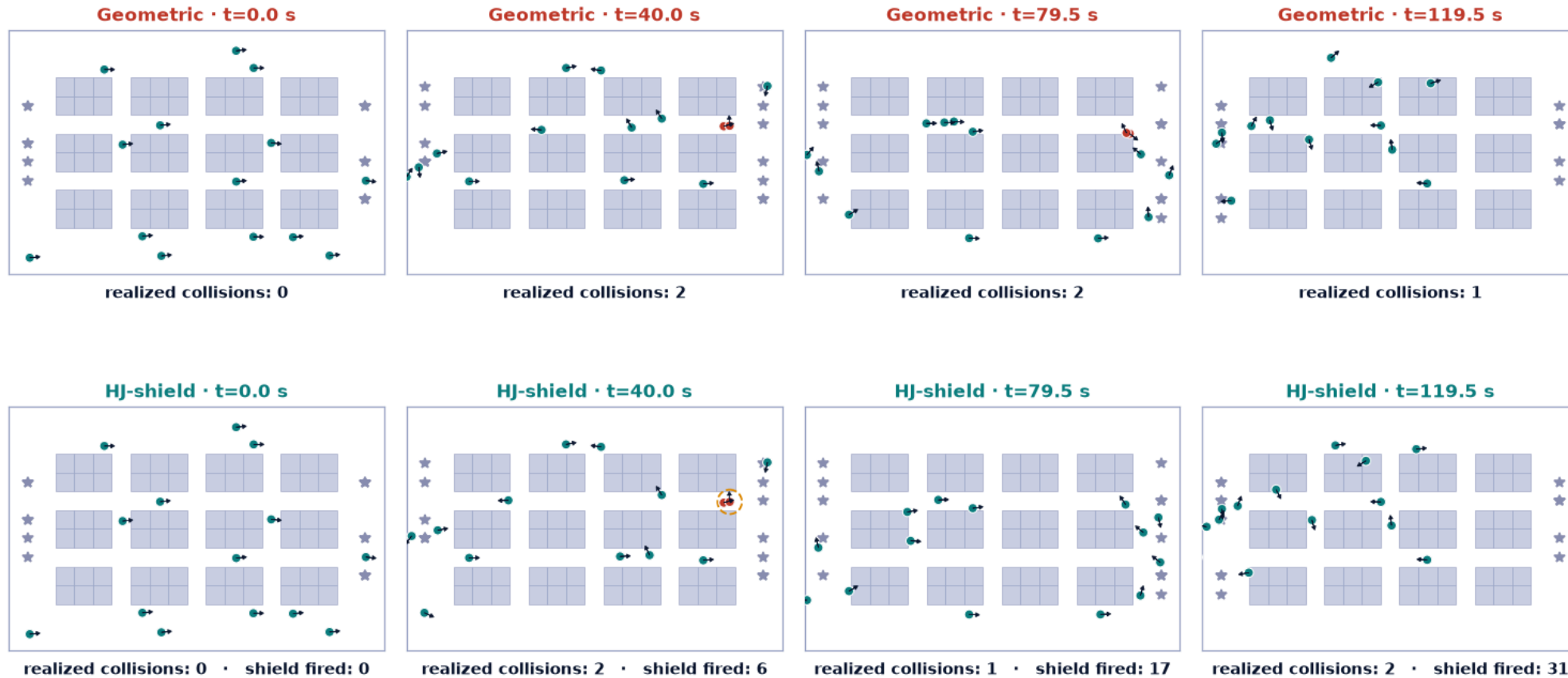
HJ-Gauss windowed-BRT shield (certified)



realized collisions: 0 · shield fired: 0

Left: geometric conflict check. Right: the HJ-Gauss windowed-BRT shield. Amber ring = shield firing (agent fleeing), red = realized contact, stars = current goals. Seed 1028 of the 30-seed harness: **98** → **66** collisions at identical throughput (0.825). Animated in the HTML deck; the PDF shows the opening frame.

▶ The Same Episode, as a Filmstrip



Four instants from the run above: top row geometric, bottom row HJ-shield. Counters under each panel accumulate realized collisions and shield activations. Identical spawn, task, and disturbance streams throughout (CRN).



The Key Scientific Finding (Why the Naive Version Fails)

- **Planning-only** BRT predicate made things **worse** (HJ 84 vs geometric 53 collisions in the first trial).
Two principled reasons:
 - i. the pursuit-evasion BRT is a **worst-case adversarial** set, but agents are **cooperative** — they never apply the BRT's safe control, so plan-time membership does not predict cooperative-tracking collisions;
 - ii. plan-time checks see only **discrete nodes at integer times**; collisions happen in **continuous time under disturbance**.
- **The certificate only bites when *actuated***. Fix: use the windowed BRT as a **runtime shield** — the lower-priority agent in an in-BRT pair **flees** (a braking "spin-in-place" bug had it get hit).

This is the arc: a negative result diagnosed to a principle, then the correct mechanism. Actuation, not inspection.



The Multiseed Statistical Harness

Mirrors the WIP-forecast harness; numpy-only:

- **≥ 30 seeds** (configurable);
- **MSER-5** warm-up truncation of the per-tick collision series (+ Welch running-mean cross-check);
- **Bootstrap 95% CIs** on every headline metric;
- **Common Random Numbers (CRN)** across policies (spawn/task/disturbance streams keyed to the seed) for a **paired** comparison;
- **Paired bootstrap CI** for $\text{mean}(h_j - \text{geom})$ + **Holm-Bonferroni** step-down FWER control across the HB family.

Single-seed results were noisy and even inverted; the harness is what turns a direction into a **defensible** claim.

Results: 30 CRN-Paired Seeds (14 Agents)

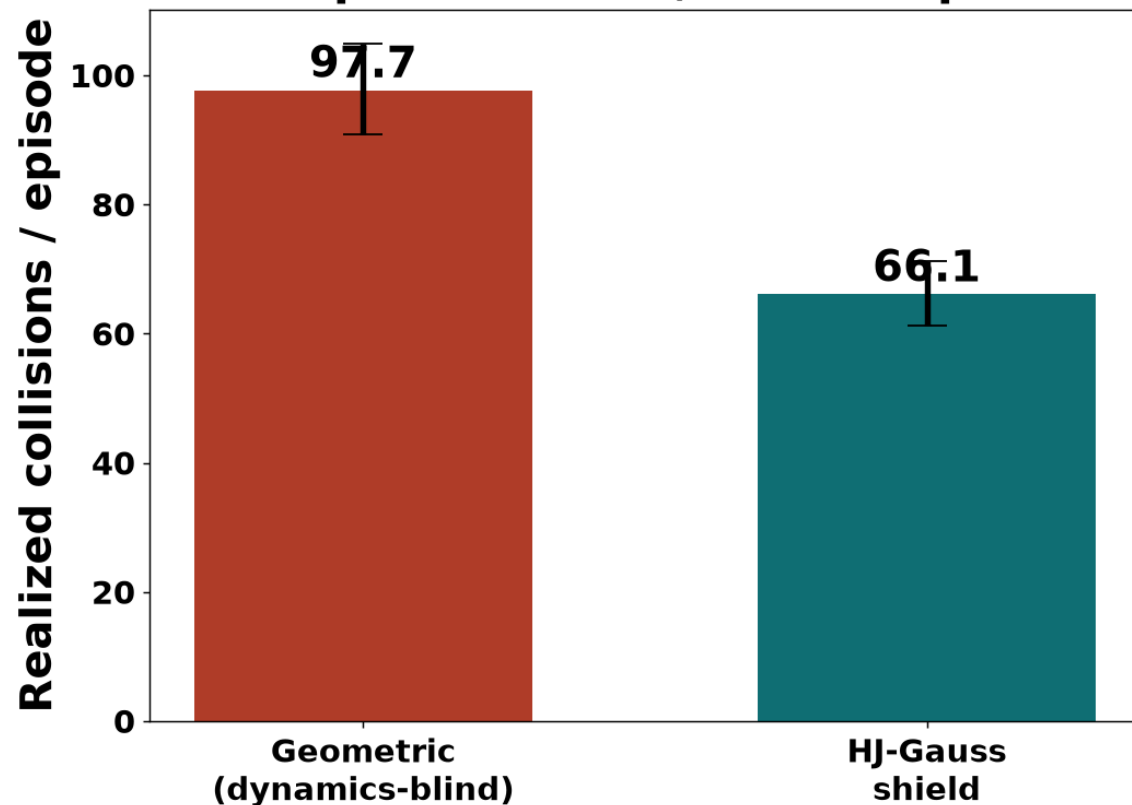
Metric	Geometric	HJ-shield	Paired diff hj-geom (Holm)
Realized collisions	97.7 [90.8, 104.9]	66.1 [61.2, 71.2]	-31.6 [-38.0, -25.4] , reject
Collision rate (MSER-5)	2.48 /tick	1.67 /tick	-0.81 [-0.98, -0.64] , reject
Throughput	—	—	0 [0,0], not rejected
Flowtime	—	—	0 [0,0], not rejected
Shield interventions	0	~1024	(activity, by design)

- **~32% fewer realized collisions**, significant under Holm-Bonferroni, at **no throughput/flowtime cost** in this model.
- Aggregated sweeps: 18-37% reduction across densities and disturbance levels.

HB-1/HB-2 rejected in the safe direction; HB-3/HB-4 show no throughput penalty. The certificate, actuated, works.

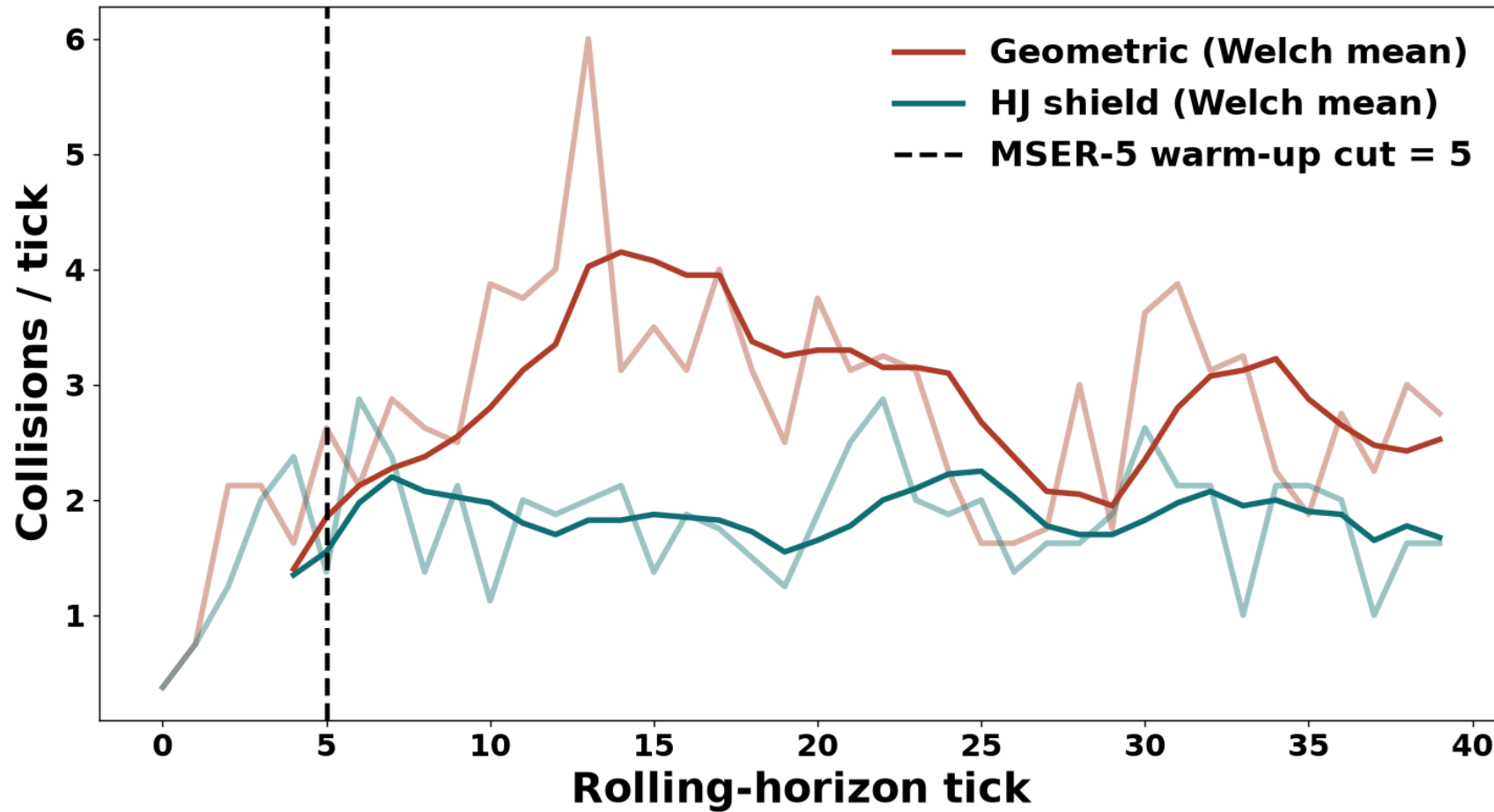
Headline: ~32% Fewer Realized Collisions

HJ shield cuts realized collisions ~32%
(30 CRN-paired seeds, bootstrap 95% CI)



30 CRN-paired seeds, bootstrap 95% CI. Geometric 97.7 → HJ-shield 66.1; paired diff **-31.6 [-38.0, -25.4]**, Holm $p \approx 0$.

Per-Tick Collisions & MSER-5 Warm-Up

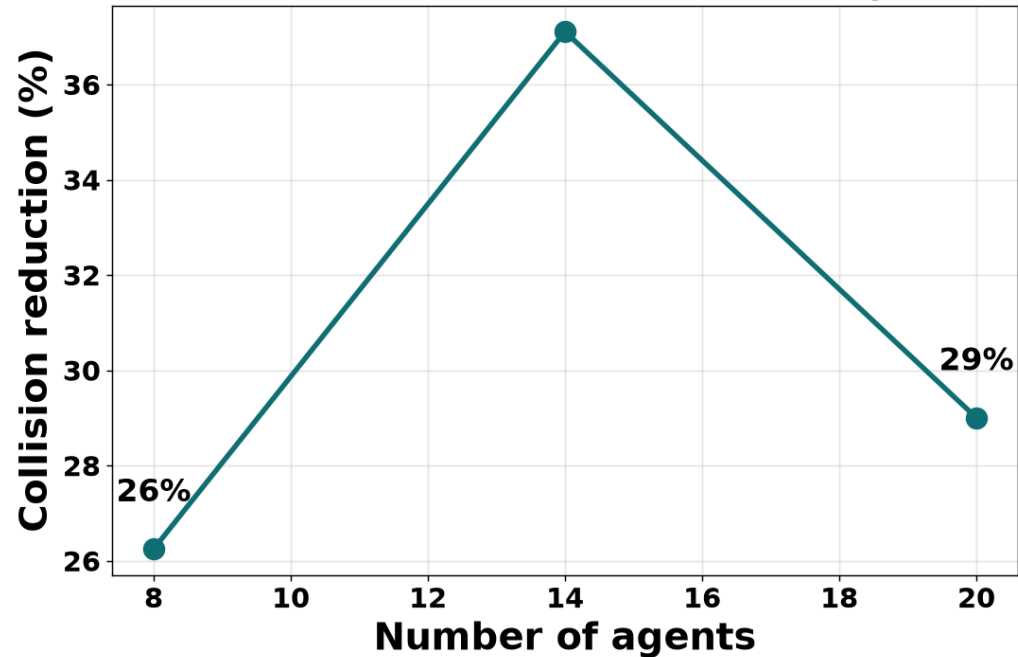


HJ shield (teal) sits below geometric (red) at every steady-state tick; the dashed line is the MSER-5 warm-up cut discarded before averaging.



The Effect Holds as the Fleet Grows

HJ shield collision reduction vs fleet size (4 seeds/point)



Collision reduction persists across agent densities (8-20 agents) — the certificate's value does not wash out with congestion.



Verdict Table

ID	Prediction	Result	Verdict
HB-1	Shield is sound	HJ never certifies a colliding plan safe; collisions ↓	✓
HB-2	Dynamics-aware advantage	31.6 fewer collisions per run, paired, Holm $p \approx 0$	✓
HB-3	$O(N \cdot n)$ memory	7.2 MB at $n = 45$; flat in resolution	✓ (paper)
HB-4	Little throughput cost	Throughput/flowtime diff = 0 in this model	✓ (with caveat)
HB-5	Fast enough	BRT precompute ~10 s; $O(1)$ online lookup	✓
HB-6	Diffusion-guidance helps	Deferred to Option C	⌚

HB-4's zero difference is partly a modeling artifact (next: Dirty Laundry). Everything else holds under the harness.

Part 7

The Diffusion Connection

The Cole-Hopf kernel *is* the diffusion kernel.

C A Mathematical Identity, Not an Analogy

- The Cole-Hopf step turns the HJ PDE into the **heat equation**, whose fundamental solution is the **Gaussian transition kernel** — the identical kernel that **score-based generative models** (DDPM, score matching) and **diffusion policies** use in their forward process.
- Under $v = -\delta \log \phi$, the recovered value gradient is a **score function**:

$$Dv \propto \nabla \log \phi = \nabla \log p.$$

- Our Feynman-Kac gradient estimator (weighted-mean shift) is structurally a **training-free score estimator**.

Certified reachability and diffusion trajectory generation **meet at the same Gaussian kernel**. This is not a metaphor; it is the same PDE.

Consequence: A Certified Guidance Term

- Diffusion trajectory planners (Diffuser, Decision Diffuser, Motion Planning Diffusion) generate paths by **denoising** but carry **no hard safety guarantee**.
- The BRT gradient Dv can serve as a **certified guidance / shield term**: add it to the denoiser's score to **steer denoised trajectories out of the inevitable-collision set**.
- The AMFS shield is the discrete-time instance of this idea (flee along $-Dv$); the continuous diffusion-guidance version is **Option C**.

A reachability-certified **shield for neural planners** — the missing safety guarantee in the learning-based MAPF line (imitation, guidance-graph, MAPF-GPT).

The Careful Claim (for the Q&A Skeptic)

- $Dv \propto \nabla \log \phi$ is a **genuine score of the Cole-Hopf-transformed value density** ϕ .
- **But** ϕ is **not** the data distribution a diffusion planner learns — the objects coincide *structurally* (same kernel, same $\nabla \log$ form), not *identically*.
- The rigorous joint-work statement: use the **certified** reachability score as a **guidance/projection** term on the learned generative score, inheriting the BRT's worst-case guarantee as a shield while the learned score supplies task performance.

State it carefully and it is bulletproof; overstate it and a diffusion expert will (rightly) push back.
Structural identity + guidance role, not "our value function is your data score."

C Joint-Work Surface with a Diffusion-Planner Team

- **Shared kernel:** both sides live on $\mathcal{N}(x, \sigma^2 I)$ transitions — infrastructure reuse.
- **Certified shield:** wrap any learned proposer (BC, CQL/IQL, diffusion) with the windowed-BRT flee/guidance term; report blocked actions and throughput margin separately.
- **Score as drift:** the importance-sampling tilt $\theta^* = -\sigma^2 c Dv$ is literally a score-guided proposal — a learned score could serve as the sampler's drift and vice versa.
- **Benchmark:** extend League-of-Robot-Runners / RHCR with a disturbance model → a **robust-throughput** axis.

One decision contract, two evidence environments: certified reachability and learned generation, meeting at the Gaussian kernel.

Part 8

Dirty Laundry

Limits, caveats, and boundaries.



Theoretical Limits

- **Quasi-linearization residual $\mathcal{E}_{\text{ql}}$** floors accuracy for non-quadratic H ; the scheme is **closed-form only when $H = \frac{1}{2}|p|^2$** . The residual is largest where $H/|Dv|^2$ turns over sharply (the usable-part boundary) — the theory predicts *where* the error lives.
- **Worst-case sample complexity is exponential in $1/\delta$** (Hoeffding + crude μ bound: $\sim e^{2(g_{\max}-g_{\min})/\delta}$), while viscosity error is only $O(\sqrt{\delta})$ — a genuine bias-variance tension. Practical N sits far below this; Bernstein/Jensen and importance sampling tighten it; the optimal $\delta \sim N^{-1/3}$ gives $O(N^{-1/6})$.
- **Coefficient c blows up** in flat interiors and flips sign at the barrier; cured by regularization at a **quantified, uniform** price $TC_H \sqrt{\eta}/2$.

None of these are hidden — each has a theorem, a remark, and a mitigation. The certificate is correct by construction (abstention band).



Reporting & Accuracy Caveats

- The Crandall-Lions $O(\sqrt{\delta})$ bound and the reported L^2 / L^∞ errors are in **different currencies** (inviscid-vs-viscous sup-norm vs MC-vs-grid); compared with explicit care.
- **Averaging $\neq$ agreement**: even the 30-seed-averaged field differs from the grid (paired Wilcoxon, $p < 10^{-8}$) — the systematic residual remains.
- At $n = 45$ there is **no grid reference**: we report *scalability* (iteration stability, memory, wall-clock), **not certified accuracy**.
- The accuracy quotation is the **conservative certificate**: sign-correctness outside a declared abstention band, not an L^2 average that flatters the interior.

L^2 flatters; L^∞ is dominated by the band we already declare undetermined. We quote the certificate, not the flattering average.



MAPF-Application Caveats

- **Adversarial vs cooperative:** the BRT is a worst-case set; using it as a cooperative-MAPF predicate is **safe but conservative**, and only reduces realized collisions when **actuated** (shield), not merely checked at plan time — this was our diagnosed negative result.
- **"Free safety" is a modeling artifact:** in the AMFS executor the discrete planner is authoritative for task progress, so shield-induced fleeing does **not** delay arrivals → HB-3/HB-4 diffs are 0. A **physical** executor should let fleeing delay arrivals, introducing a small, real throughput cost.
- **Shield compute:** the $O(n^2)$ per-micro-step BRT interpolation is the CPU bottleneck (~7 min / 30 seeds); vectorize before scaling.
- **Scope:** Dubins agents, one floor topology, single disturbance level — a PoC, not a fleet study.

The result is real and significant; the *magnitude* and cost need a more physical executor and larger sweeps (Option C, GPU).



Source & Scope Boundaries

- Grid solvers remain **preferable for $n \leq 4$** (smaller constant, no sampling variance); HJ-Gauss is the method of choice only in the high-dimensional regime. The paradigms are **complementary**.
- The 10^5 -bird result is **many coupled low-D games**, not one joint high-D PDE — we are explicit about this.
- All experiments are **single-CPU**; GPU/Lambda scaling (Part 9) is future work, not claimed here.
- The diffusion connection is a **structural identity + guidance role**, not a claim that our value density equals a learned data distribution.

Complementary, decoupled, single-CPU, structural — four boundaries stated plainly so no claim is overread.

Part 9

Conclusions, Future Work, References

🚩 Conclusions

- **HJ-Gauss** turns the nonlinear viscous HJ PDE into a **sequence of linear heat equations** (Cole-Hopf), solved as **Gaussian Feynman-Kac expectations** by a frozen-coefficient **Picard iteration (Algorithm 1)**.
- Memory drops from grid $O(M^n)$ to $O(N \cdot n)$ — storage- and discretization-free.
- **Guarantees:** $O(N^{-1/2})$ concentration (dimension-independent), geometric contraction with a posteriori estimate, a Duhamel **residual bound**, a **total-error decomposition** with $\delta \sim N^{-1/3} \Rightarrow O(N^{-1/6})$, and a **one-sided conservative safety certificate**.
- **Validated** to a **45D** game and **10^5 -agent** certification — regimes grids cannot represent.
- **For autonomous mobility:** a dynamics-aware, disturbance-robust **conflict predicate/shield** for MAPF (windowed BRT = RHCR window), and — via the Gaussian/score identity — a **certified guidance term for diffusion planners**.

Certified safety is the sign of a value function. HJ-Gauss computes that sign where grids cannot — and hands the same object to your learned planners.

🚩 Future Work

- **Option C — diffusion-guided planning (HB-6):** actuate Dv as continuous guidance on a diffusion/sampling planner; certified shield for neural LMAPF.
- **Physical MAPF executor:** let shield fleeing delay arrivals → measure the *real* throughput-vs-safety frontier (robust-throughput benchmark).
- **GPU / Lambda Labs scaling:** vectorize the shield's BRT interpolation; run the harness as a Metaflow flow (fan-out over configs × seeds); scale the BRT precompute (bigger grids, higher-D dynamics) on H100s.
- **Adaptive importance sampling** in high dimensions; Bernstein-tight sample budgets.
- **Tighter integration** with safe-RL / policy-certification pipelines.

The scaling path (`imp` /ar-metaflow, W&B, S3 checkpointing) is scoped in `AMFS/infra/lambda_labs_notes.md`.

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Code: github.com/robotsorcerer/levelsetpy ([monte_carlo](#), [monte_carlo/AMFS](#)) · Contact: ogunmolu@amazon.com

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Thank You

Questions?



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Certified safety is the sign of a value function —
HJ-Gauss computes that sign where grids cannot.

Backup slides: full proofs (residual Duhamel bound, contraction), murmuration topology theorems, AMFS harness internals, Lambda Labs scaling plan.

Appendix

Background & Foundations (Parts A–C)

Hamilton-Jacobi and viscosity theory (A), reachability foundations (B), and the LevelSetPy grid pipeline (C) — reference material for the main talk.



Who Should Care, and Why

- **Autonomous mobility / MAPF:** a certified, dynamics-aware conflict predicate that drops into rolling-horizon planners at fleet scale.
- **Safe RL / policy verification:** backward tubes for a closed-loop learned policy are memory-bound on grids; $O(N \cdot n)$ lifts that bound.
- **Differential games / pursuit-evasion:** the adversarial worst-case reachable set is what HJ-Isaacs computes.

The unifying object is a **value function whose sign certifies safety**. This talk is about computing that sign where grids cannot.



Who Should Care, and Why

- **Differential games / pursuit-evasion:** the adversarial worst-case reachable set is what HJ-Isaacs computes.
- **Diffusion / generative planning:** the Cole-Hopf kernel *is* the score-based diffusion kernel; the value gradient is a score.
- **Air-traffic, collision avoidance, multi-robot control:** the classical application domain of HJ reachability.

The unifying object is a **value function whose sign certifies safety**. This talk is about computing that sign where grids cannot.

Part A

Hamilton-Jacobi PDE Theory Recap

Viscosity solutions, vanishing viscosity, and why we need them.



The Cauchy-Type HJ Equation

Our chief object is the evolution (Cauchy) HJ equation

$$v_t(x, t) + H(t; x, \nabla_x v) = 0 \quad \text{in } \Omega \times (0, T], \quad v(x, 0) = v_0(x) \quad \text{in } \Omega,$$

with $H : (0, T] \times \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ continuous and g, v_0 **bounded, uniformly continuous (BUC)**.

- A closely related object is the **scalar conservation law / convection equation** $v_t + \sum_i f_i(v)_{x_i} = 0$, whose shock theory motivates the numerics.
- The HJ equation is **first-order, nonlinear, hyperbolic**.

First-order nonlinear hyperbolic PDEs generically fail to have smooth global solutions — the reason the entire viscosity-solution apparatus exists.



Why Classical Solutions Fail: Crossing Characteristics

- The **method of characteristics** integrates the PDE along curves in (x, t) .
- For nonlinear H , characteristics **cross**: multiple characteristics carry conflicting values to the same point $\rightarrow$ a **shock** / gradient discontinuity.
- Consequence: no global C^1 solution exists in general, **even when H and g are smooth**.
- Global analysis via classical PDE theory is "virtually impossible" (Crandall-Lions).

The value function of a differential game develops **kinks and shocks** right at the barrier — the most safety-relevant region. We must define a *weak* solution that survives there.



Viscosity Solutions¹

- **Viscosity solutions** are the correct weak solution class for HJ PDEs: they exist almost everywhere, and enjoy **existence, uniqueness, and stability** theorems.
- Definition (sketch) via test functions: v is a viscosity *subsolution* if for every smooth φ touching v from above at (x_0, t_0) , $\varphi_t + H(\cdot, \nabla \varphi) \leq 0$; *supersolution* with the reverse inequality and touching from below; a **solution** is both.
- This selects the physically-correct kinked solution and discards spurious ones.

Viscosity theory gives us a **unique** object to compute and to certify — indispensable for a safety guarantee.

¹ Crandall & Lions, 1983



Vanishing Viscosity

Introduce $\delta > 0$ and regularize (parabolic smoothing, as in gas dynamics):

$$v_t^\delta + H(t; x, \nabla v^\delta) = \frac{\delta}{2} \Delta v^\delta \quad \text{in } \Omega \times (0, T], \quad v^\delta(x, 0) = g(x).$$

- The added Laplacian makes the PDE **parabolic**, hence classically well-posed: $v^\delta \in C^{2,1}$, unique, stable, BUC for all T .
- Traversing the limit $\delta \rightarrow 0$ recovers the unique viscosity solution.

δ is the linchpin of HJ-Gauss: it is both the **smoothing** that admits a classical solution *and* the **diffusion coefficient** of the heat equation we are about to expose.



The Crandall-Lions Error Bound

The viscous solution approximates the inviscid viscosity solution at a known rate:

$$\sup_{t \in (0, T]} \sup_{x \in \mathbb{R}^n} |v(t, x) - v^\delta(t, x)| \leq k\sqrt{\delta}, \quad k > 0.$$

- **Bias scales as $O(\sqrt{\delta})$:** smaller $\delta \rightarrow$ more accurate.
- But (preview) smaller $\delta \rightarrow$ more concentrated exponential weights $\rightarrow$ higher Monte-Carlo variance.
- This is the **bias-variance knob** we will formalize in Part 4 ($\delta \sim N^{-1/3}$).

One scalar δ trades geometric accuracy against sampling variance. Choosing it well is the practical art of the method.



Admissible Data on $\mathbb{R}^n$

- Whole-space heat theory requires the datum in $C(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$.
- The transformed datum is $\omega^\delta(0, \cdot) = e^{-c g}$ — **not** g itself. This matters: the signed distance ℓ is *unbounded above*, but for a bounded target $g \geq g_{\min}$ and $g \rightarrow +\infty$ as $|x| \rightarrow \infty$, so

$$0 < e^{-c g(y)} \leq e^{-c g_{\min}}, \quad e^{-c g(y)} \rightarrow 0 \text{ as } |y| \rightarrow \infty.$$

- Hence $e^{-c g}$ is continuous, strictly positive, bounded, decaying $\rightarrow$ the heat solution is the **unique bounded** one, the integral converges absolutely, and the Gaussian expectation is genuine and **unbiased**.

Only the **lower** bound $g \geq g_{\min}$ is needed; no upper bound on g is used. This is what lets us sample over all of $\mathbb{R}^n$.

Part B

Reachability & Safety Foundations

What a certified safety set *is*, before we compute one.

What Is Reachability?

- **Reachability** concerns the *decidability* of a dynamical system's trajectory evolution across a phase space.
- A reachable system is **decidable** when one can compute *all* states reachable from an initial condition **in a finite number of steps**.
- Dual questions:
 - **Forward**: where can the system go from here?
 - **Backward**: from which states is a target inevitably reached (or avoidable)?
- Safety analysis is naturally **backward**: characterize the set of states doomed to enter a danger set, then stay out of it.

Certifying a learned controller, neural policy, or planner means proving it satisfies all specified requirements — a **verification** problem. Reachability is the geometric engine of that verification.

Backward Reachable Sets and Tubes

- **Backward Reachable Set (BRS)**: states that reach the target at a *specific* time.
- **Backward Reachable Tube (BRT)**: states that reach the target at *some* time in $[0, T]$ — the safety-relevant object.
- **Reach-Avoid Tube (BRAT)**: states that can reach a goal while avoiding an obstacle set.
- **Robustly-Controlled BRT (RCBRT)**: when the controller must counter a **worst-case disturbance** — the adversarial guarantee.

$$\mathcal{L}([-T, 0], \mathcal{L}_0) = \{x \in \mathbb{R}^n : \exists \beta \in \mathcal{B}(t) \forall u \in \mathcal{U}(t), \exists \bar{\tau} \in [-T, 0], \xi(\bar{\tau}) \in \mathcal{L}_0\}.$$

The strategy structure $(\exists \beta \forall u)$ encodes the game: disturbance plays a nonanticipative strategy β against control u .

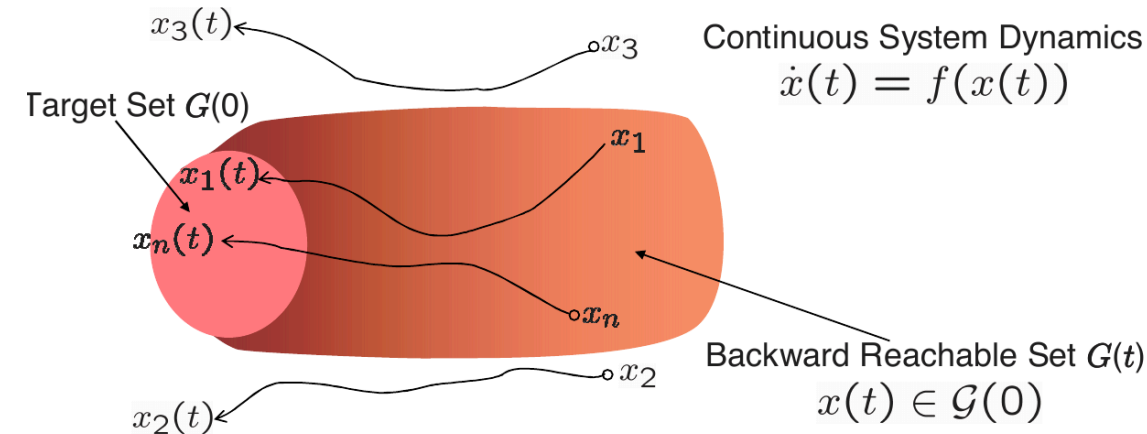


Why "Backward" Reachable Sets

- A **continuous backward reachable set** is the set of all states from which trajectories can reach a given target set $G(0)$.
- Called "**backward**" to distinguish it from the *forward* reachable set.
- **To compute it, run the dynamics backward in time** from the target set.

For safety, the target is usually the *unsafe* set, so the backward reachable set is the set of states doomed to become unsafe.

(Mitchell 2004, Reach Sets and the HJ Equation)

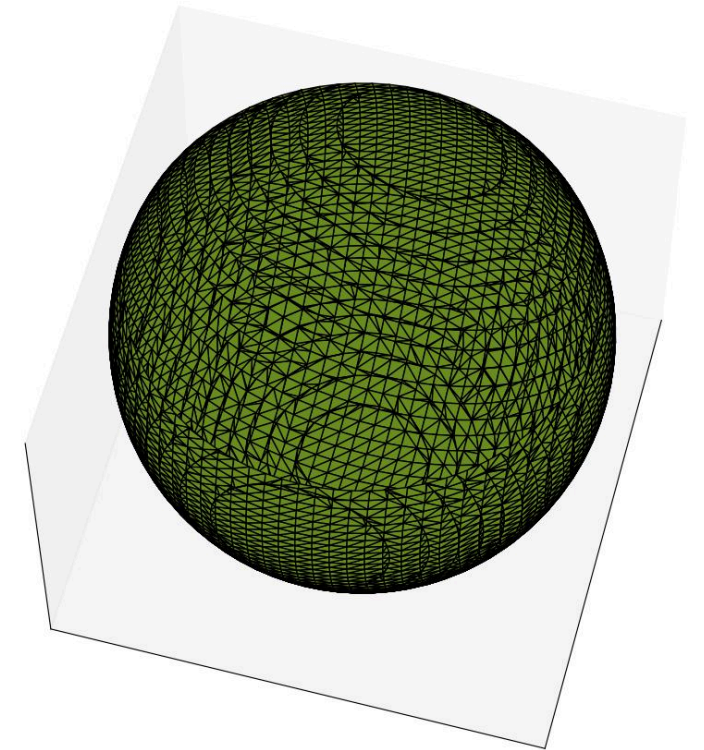


🎓 How to *Represent* the Set/Tube?

Computing a reachable set poses two coupled problems:

- **Represent** the set of reachable states.
- **Evolve** that set according to the dynamics.

Inset: A sphere as an SDF — LevelSetPy (L. Molu), ACM TOMS 51(2), 2025.



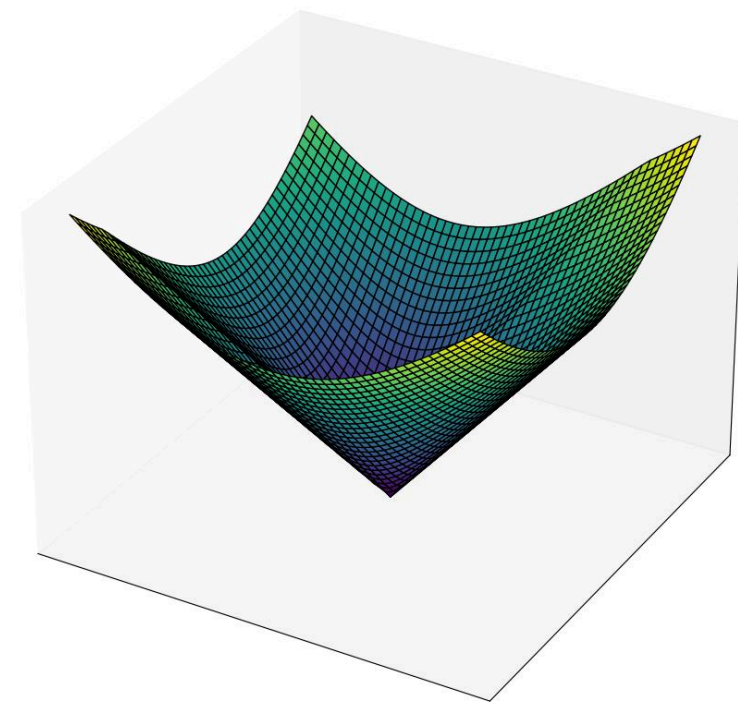
Implicit Surface Functions

Level-set idea: Represent a set $G(t)$ as an **isosurface of a scalar function** $\phi(x, t)$ ¹:

- **State-space dimension does not matter conceptually**: the same ϕ machinery works in any n .
- Surfaces **automatically merge and separate** as the set evolves.
- **Geometric quantities** (normals, curvature, distance) are easy to compute from ϕ .

Critical talk juncture: the reachable set is the zero sublevel set of a value function ϕ .

Inset: Set union of two spheres — LevelSetPy (L. Molu), ACM TOMS 51(2), 2025. · ¹ Osher & Sethian



How to *Represent* the Set/Tube?

For continuous systems $\dot{x} = f(x)$, two families:

- **Lagrangian** (forward sets; restricted dynamics/shapes; overapproximation): HyTech, Checkmate, d/dt , ellipsoidal¹.
- **Eulerian** (backward sets; general dynamics incl. competitive inputs; **implicit** set representation).
 - **HJ-Gauss** belongs here.

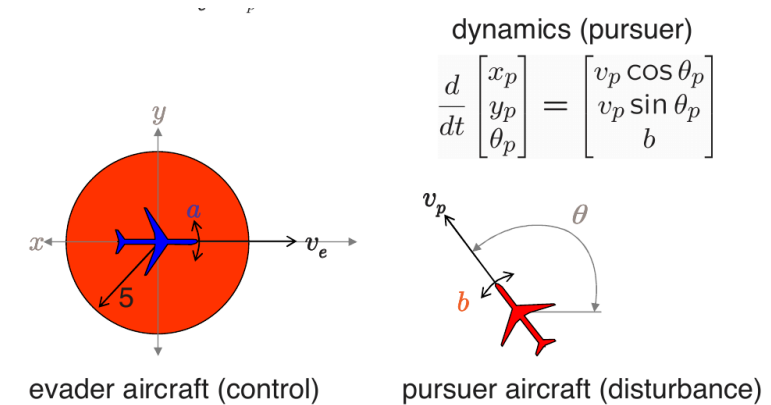
¹ Kurzhanski

Canonical Example: Two Identical Vehicles

Classical collision avoidance:

- Collision if the vehicles come within **5 units** of each other.
- **Evader** picks turn rate $|a| \leq 1$ to *avoid*;
- **Pursuer** picks $|b| \leq 1$ to *cause* collision; fixed equal speeds $v_e = v_p = 5$.

(Mitchell 2004, Reach Sets and the HJ Equation)



24 Oct 04

Ian Mitchell (UBC Computer Science)

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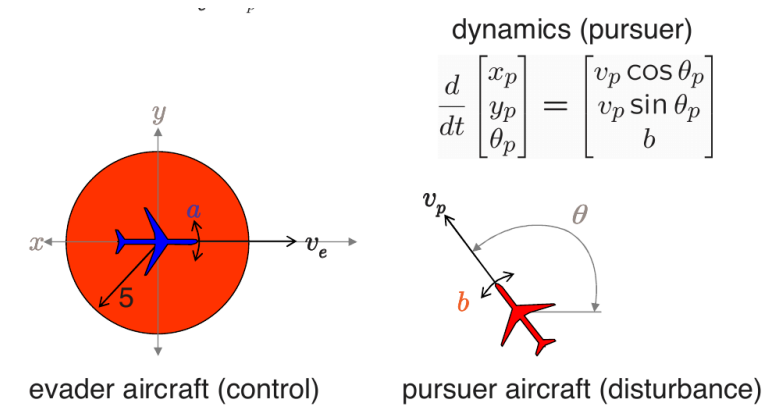
Canonical Example: Two Identical Vehicles

Classical collision avoidance:

- Work in **relative coordinates** with the evader fixed at the origin.
- State is relative position (x, y) and relative heading ψ .

NB: Same relative-coordinate pursuit-evasion (PE) game we solve with HJ-Gauss (Part 5); and the pairwise MAPF conflict of Part 6.

(Mitchell 2004, Reach Sets and the HJ Equation)



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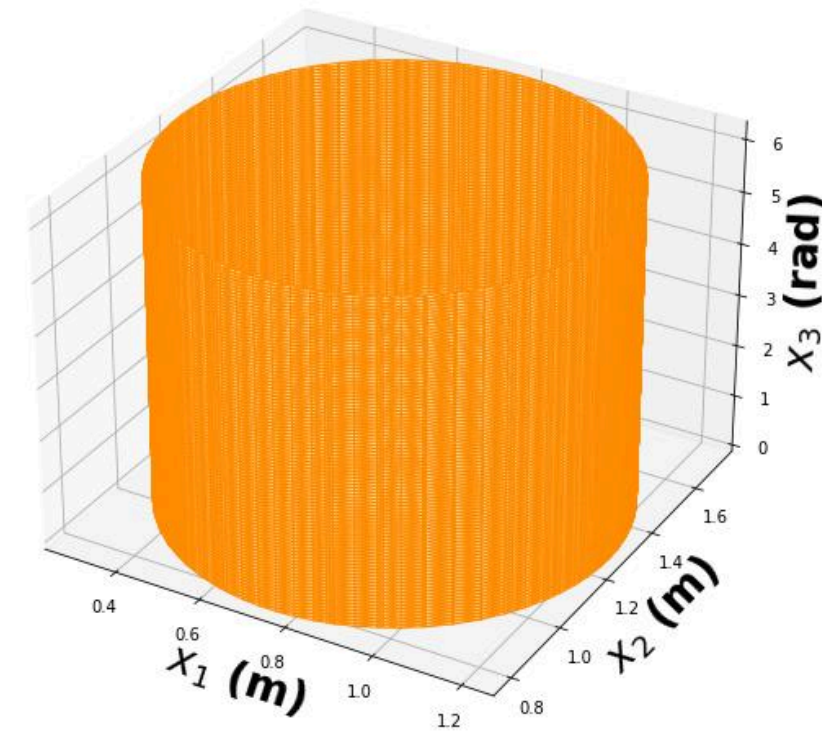
Evolve: The Time-Dependent HJ Equation

The set evolves by a (modified) Hamilton-Jacobi PDE:

- A **first-order hyperbolic PDE** whose solution can form **kinks** (discontinuous derivatives).
- The right weak solution is the **viscosity solution**¹.

(Mitchell 2004, Reach Sets and the HJ Equation); viscosity:
Crandall-Evans-Lions; level sets: Osher-Sethian · Dubins BRT:
LevelSetPy (L. Molu), ACM TOMS 51(2), 2025 · ¹ Crandall, Evans,
Lions

BRT at 0.00 secs.



Evolve: The Time-Dependent HJ Equation

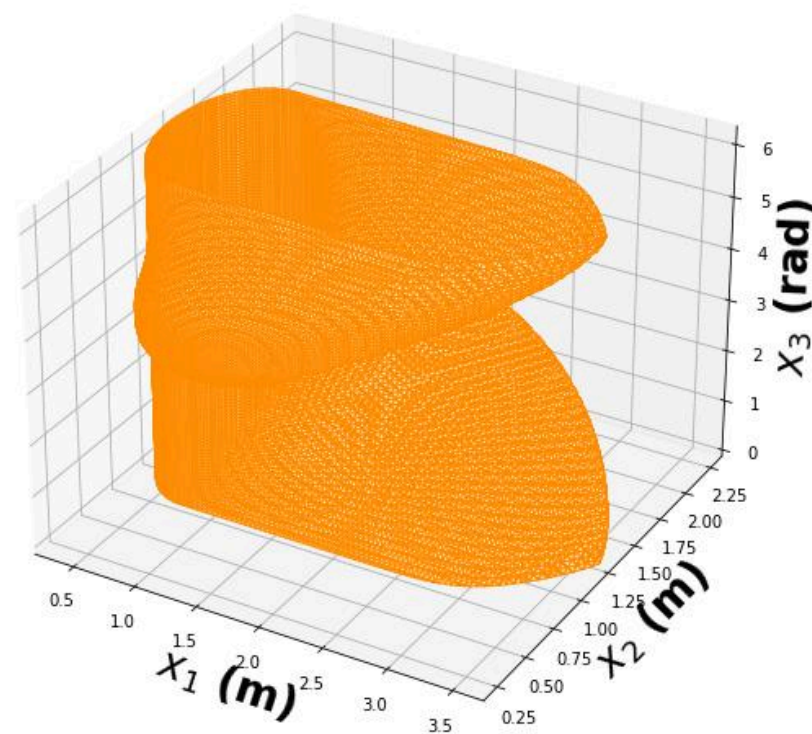
The set evolves by a (modified) Hamilton-Jacobi PDE:

- **Level-set methods** produce convergent numerical schemes¹:
 - Non-oscillatory high-accuracy spatial derivatives;
 - A stable/consistent numerical Hamiltonian; and
 - Total variation-diminishing (TVD) high-order explicit time integration.

This is the classical grid pipeline of Part C. HJ-Gauss tackles this computational cost.

(Mitchell 2004, Reach Sets and the HJ Equation); viscosity:
Crandall-Evans-Lions; level sets: Osher-Sethian · Dubins BRT:
LevelSetPy (L. Molu), ACM TOMS 51(2), 2025 · ¹ Osher, Sethian

BRT at 2.50 secs.



The Game Value and the Optimal-Stopping Fix

- The **terminal-cost differential game**:
 - Trajectories $\xi(\cdot; x, t, a, b)$;
 - **Value function** $\phi(x, t)$ is the viscosity solution of the HJ equation¹.

This modified/augmented Hamiltonian is the $\min\{0, \cdot\}$ freeze term on our HJI-RCBRT slide.

¹ Evans & Souganidis, 1984

The Game Value and the Optimal-Stopping Fix

- To stop trajectories from passing *through* the target $G(0)$ (so the set is a **tube**, not just a set),
 - **Augment the disturbance input;**
 - The augmented HJ equation solves for the reachable set;
 - The augmented Hamiltonian is the modified $\min\{0, H\}$ Hamiltonian.

This modified/augmented Hamiltonian is the $\min\{0, \cdot\}$ freeze term on our HJI-RCBRT slide.

Three Eulerian Approaches (All Equivalent)

The method sits among Eulerian formulations:

- **Static HJ**: Minimum time-to-reach¹; (dis)continuous implicit representation; yields optimal-input information.
- **Viability kernels**: Set-valued analysis for very general dynamics²; discrete implicit representation; overapproximation guarantee.
- **Time-dependent HJ** (this method): Continuous solution, optimal-input information throughout the state space, high-order accurate.

All three are **theoretically equivalent**; HJ-Gauss is a new solver for the time-dependent HJ formulation.

(Mitchell 2004, Reach Sets and the HJ Equation); Falcone/Sethian; Aubin/Saint-Pierre · ¹ Falcone; Sethian · ² Aubin; Saint-Pierre

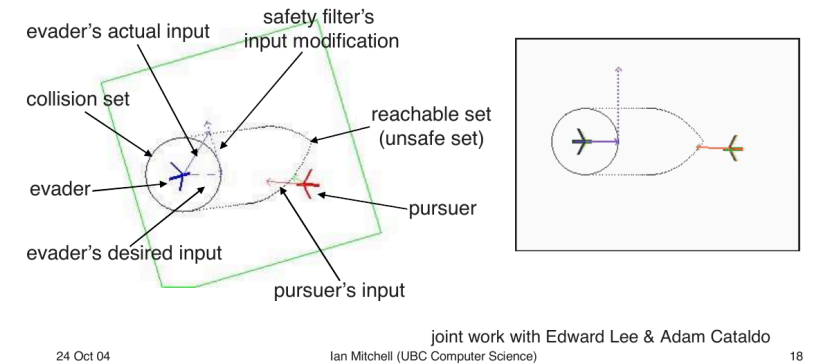
🎓 Why It Matters: Two Applications

Reachable sets already drive real-world safety systems:

- **Softwalls for aircraft safety:** Filter evader's input so the pursuer never enters the reachable (unsafe) set — a certified safety filter (with E. Lee & A. Cataldo).
- **Collision alert for ATC:** Flag aircraft pairs whose flight plans intersect and whose relative state enters the collision region.
 - A one-hour Oakland-airspace sample: **1590 pairs, 25 detected conflicts, 2 false alerts.**

These are the ancestors of the certified conflict predicate / shield we bring to warehouse MAPF in Part 6.

(Mitchell 2004, Reach Sets and the HJ Equation); Softwalls with E. Lee & A. Cataldo





Validation: Merz's Analytic Solution

- For **identical PE** dynamics, the reachable set admits an **analytic solution**¹.
- Merz placed the pursuer at the origin; the game is **not symmetric**.
- That analytic solution is used to **validate the numerical algorithm**² — the same discipline behind our LevelSetPy-vs-Monte-Carlo comparison in Part 5.

Takeaway of the primer: reachability = represent implicitly, evolve in a HJ PDE game, solve for the viscosity solution.

HJ-Gauss keeps this exactly and changes only *how* the PDE is solved.

(Mitchell 2004, Reach Sets and the HJ Equation); Merz (1972); Mitchell (2001) · ¹ Merz, 1972 · ² Mitchell, 2001

Target Set and Zero Level Set

- The **target set** at horizon T is the invariant set

$$\mathcal{L}_0(T) = \{x \in \mathbb{R}^n : v(0, x) \leq 0\},$$

robustly controlled over the **distance-to-target** cost $g(0, x)$.

- Numerically g is a **signed-distance function** $\ell(x)$ whose zero sublevel set is the target: negative inside, positive outside.
- The BRT is recovered as the **zero sublevel set** of the value function:

$$x(t) \in \mathcal{L}(\cdot) \iff v(t, x) \leq 0.$$

The Hamilton-Jacobi-Isaacs Value Function

The value function of the RCBRT is the **viscosity solution** of the variational HJ-Isaacs equation

$$v_t(t, x) + \min\{0, H(t; x, Dv)\} = 0, \quad v(0, x) = g(0, x),$$

with the **game Hamiltonian**

$$H(t; x, p) = \max_u \min_w \langle p, f(t; x, u, w) \rangle.$$

- The $\min\{0, \cdot\}$ **freeze** term makes the set only grow (a tube, not a set) — states already captured stay captured.
- For reach-avoid, the variational inequality couples the growth term with the obstacle datum ℓ .

This Hamiltonian is **positively 1-homogeneous in p , state-dependent, nonconvex, and sign-changing** across the barrier. Those four properties will rule out every convex-duality shortcut.

Reach vs Reach-Avoid; Sign & Time Conventions

- **Backward-reachability convention:** t is the **backward horizon** (time-to-go); the datum is posed at $t = 0$ and the tube grows over $(0, T]$.
- **Reach (capture) tube:** $v_t + \min\{0, H\} = 0$.
- **Reach-avoid tube (viscous):**

$$\min \left\{ v_t^\delta + H^\delta - \frac{\delta}{2} \Delta v^\delta, \quad g(t, x) - \ell(t, x) \right\} \leq 0.$$

- Physical time is $t_{\text{phys}} = T - t$; the datum is the terminal cost in physical time.

These conventions matter for signs during implementation and for *which* set (reach vs avoid) a negative value certifies. We adopt the backward-reachability viscosity-solution convention throughout.

Why This Is the Right Safety Object

- **Worst-case, not average-case:** the RCBRT certifies safety against *all* admissible disturbances — the guarantee a safety case needs.
- **Geometric and set-valued, not trajectory-valued:** it characterizes every unsafe initial condition at once, not one rollout.
- **Composable:** the zero level set can be intersected, unioned, and propagated; it plugs into supervisory control and shielding.
- **Certificate-grade:** a signed value with an error bound yields a decision — SAFE / UNSAFE / UNDETERMINED (Part 4).

The cost of this rigor is computational: solving the HJ(I) PDE. The rest of the talk is about paying that cost at scale.

Part C

The Grid Pipeline: LevelSetPy

How Reachable Sets Are Computed Today — and Why It Is $O(M^n)$.

Molu. ACM TOMS 2025 · IEEE CDC 2024

TOMS: ACM Transactions on Mathematical Software · CDC: Conference on Decision and Control.

The Level-Set Idea¹

- Represent the reachable set implicitly as the **zero sublevel set** of $v(t; x)$ stored on a Cartesian grid.
- Evolve the interface by integrating the HJ PDE on the grid:
 - Discretize space (upwinding),
 - Stabilize the Hamiltonian (Lax-Friedrichs),
 - March in time (TVD-RK).

¹ Osher-Sethian

The Level-Set Idea¹

- **LevelSetPy** (our prior work) reimplements the 2004 MATLAB Level Set Toolbox in NumPy/CuPy, GPU-accelerated and interoperable with modern Python (PyTorch, SciPy, ROS).

MATLAB ToolboxLS (2004)	LevelSetPy (2024/25)
CPU-only, single-threaded	NumPy + CuPy GPU
Slow for modern problems	Fast, batched, portable
No modern-library plug-in	ROS/PyTorch/SciPy compatible

LevelSetPy makes grid reachability *fast*. It does not change the *memory scaling* — the theme of Part 1.

¹ Osher-Sethian

Implicit Surfaces & Signed-Distance Initialization

- The target set is initialized as a **signed-distance function** $\ell(x)$: negative inside, positive outside, $|\nabla \ell| = 1$.
- Stored as an n -dimensional array over a Cartesian grid: M points per axis $\rightarrow M^n$ cells.
- Geometric primitives (spheres, cylinders, half-spaces) and boolean set operations (union/intersection/complement) are implemented as min/max on the level-set field.
- Example: a capture ball of radius r is $\ell(x) = \|x\| - r$; a cylinder ignores the periodic θ axis.

The implicit representation is elegant and closed under set algebra — but it stores the value at **every** grid node.

Spatial Derivatives via Upwinding

- The co-state $p = \partial v / \partial x$ must be approximated from grid values with the correct **direction of information flow** (upwinding) to remain stable at shocks.
- **First-order upwinding**: one-sided differences chosen by the sign of the characteristic speed.
- Left/right approximations D^-v , D^+v feed the numerical Hamiltonian; the choice prevents differencing across a discontinuity.

$$v_x(x, t) \approx \frac{\partial v(x, t)}{\partial x} \quad (\text{one-sided, direction by wind}).$$

Naive centered differences ring and go unstable at the barrier; upwinding is the minimal fix, refined next by ENO/WENO.

ENO: Essentially Non-Oscillatory Reconstruction

- **ENO** picks, among candidate stencils, the **smoothest** one to interpolate the derivative — avoiding stencils that straddle a shock.
- Orders implemented: **ENO2** (2nd), **ENO3** (3rd), via `upwind_first_eno2.py`, `upwind_first_eno3.py`.
- Higher order → sharper interface, lower numerical diffusion, at more stencil work per node.

ENO chooses *one* smoothest stencil. WENO improves on this by blending stencils with adaptive weights.

WENO5: Weighted ENO Shock Capture

- **WENO5**¹ forms a **convex combination** of three candidate substencils rather than choosing one, achieving 5th-order accuracy in smooth regions and non-oscillatory capture at shocks.
- Substencils on the grid index i :

$$\{i - 3, \dots, i\}, \quad \{i - 2, \dots, i + 1\}, \quad \{i - 1, \dots, i + 3\},$$

combined with nonlinear weights that de-emphasize stencils crossing a discontinuity.

- Implemented in `upwind_first_weno5.py`.

WENO5 is the workhorse for accurate reachable-set boundaries — and its per-node cost is one reason grid solves are expensive even before the memory wall.

¹ Liu, Osher, Chan 1994

The Lax-Friedrichs Numerical Hamiltonian

- The analytic $H(x, p)$ is replaced by a **monotone numerical Hamiltonian** $\hat{H}(x, p^-, p^+)$ using the Lax-Friedrichs flux:

$$\hat{H} = H\left(x, \frac{p^- + p^+}{2}\right) - \frac{1}{2} \alpha \cdot (p^+ - p^-),$$

with dissipation coefficient $\alpha = \max |\partial H / \partial p|$ over the relevant range (dimension-wise for global LF).

- Monotonicity guarantees convergence to the viscosity solution.

The artificial dissipation α is a *numerical* cousin of the viscosity δ — both stabilize the barrier. HJ-Gauss will trade the grid's LF dissipation for the analytic δ .



Time Integration: TVD Runge-Kutta

Method of lines + **Total-Variation-Diminishing Runge-Kutta**¹, so the interface does not spuriously oscillate:

- **Forward Euler (1st):** $v^{n+1} = v^n + \Delta t L(v^n), L = -\hat{H}$.
- **TVD-RK2:** Euler step to $v^{(1)}$, second Euler step, then convex average $v^{n+1} = \frac{1}{2}v^n + \frac{1}{2}v^{(2)}$.
- **TVD-RK3:** three stages with convex-combination weights $(1), (3/4, 1/4), (1/3, 2/3)$.

TVD-RK preserves monotonicity of the spatial scheme in time. Each stage is a full grid sweep.

¹ Shu-Osher



The CFL Condition

- Explicit integration is stable only under a **Courant-Friedrichs-Lewy** step restriction:

$$\Delta t \leq \frac{\text{CFL}}{\sum_i \max |\partial H / \partial p_i| / \Delta x_i}, \quad \text{CFL} \in (0, 1).$$

- Numerical information must not travel more than one grid cell per step.
- Finer grids ($\Delta x \downarrow$) force **smaller** $\Delta t \rightarrow$ more steps $\rightarrow$ compounding the cost.

CFL couples spatial and temporal resolution: refining space to sharpen the barrier makes *both* the memory and the step count worse.

Package Anatomy & a Worked Example

Module	Role
<code>grids</code> , <code>initialconditions</code>	Cartesian grids, signed-distance shapes
<code>spatialderivative</code>	Upwind first, ENO2/3, WENO5
<code>explicitintegration</code>	Lax-Friedrichs Hamiltonians, TVD-RK, CFL
<code>dynamicalsystems</code>	Vehicle models (Dubins, rockets)
<code>visualization</code>	Isosurface / marching cubes rendering

- **Worked example:** the rockets-launch pursuit-evasion game; 2D (x, z) slices of the 3D relative-state BRT, evolved backward over $(0, T]$.
- Multi-agent verification: flocks/murmurations partitioned into per-flock games.

A clean, tested, GPU-portable stack — the state of practice we now try to move beyond.

Grid Complexity: The Accounting

- **Memory:** $O(M^n)$ — store the value (and gradient) at every node.
- **Time per step:** $O(M^n)$ node updates $\times$ stencil width $\times$ RK stages, under a CFL-bounded step count.
- GPU acceleration cuts the *constant* and parallelizes node updates, but the **exponent n is untouched**.

$$n = 6, M = 100 \Rightarrow 10^{12} \text{ cells} \approx 8 \text{ TB per double array.}$$

This is the wall. Part 1 quantifies it and surveys who has tried to climb it.